REVIEWS OF GEOPHYSICS AND SPACE PHYSICS, VOL. 22, NO. 1, PAGES 47–72, FEBRUARY 1984

Paleolatitudes Determined From Paleomagnetic Data From Vertical Cores

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Most of the paleomagnetic data available from oceanic plates consist of measurements of inclination but not declination from azimuthally unoriented vertical cores. Similarly, the basic paleomagnetic data to be analyzed from crustal blocks that have rotated by varying amounts about vertical axes are a set of paleoinclinations. A new method is proposed for determining the paleocolatitude of a sampling site and the associated confidence limits. Paleomagnetic inclinations are first converted to paleocolatitudes and averaged to produce a mean colatitude. After showing that this mean is biased, we derive a paleocolatitude correction to remove the bias, using a model for geomagnetic secular variation based on the latitude dependence of secular variation during the past 5 million years. After testing the validity of the model by applying it to paleomagnetic data from Quaternary lavas for which both inclination and declination are known, we apply the method to inclination-only paleomagnetic data from Suiko Seamount, the most complete data set from a drilling site. The method is then applied to paleomagnetic data from all Deep Sea Drilling Project sites for the Pacific plate to obtain for each site an unbiased mean paleocolatitude and its confidence limits. These paleocolatitudes and confidence limits are then used as input for a least squares analysis to obtain Early and Late Cretaceous poles for the Pacific plate and 95% confidence limits for these poles. It is shown that the scatter of individual site paleocolatitudes about the best fit pole is consistent with the confidence limits at each site as determined by the new method of analysis.

INTRODUCTION

Azimuthally unoriented vertical cores comprise most of the material available for paleomagnetic measurements from many oceanic plates. The paleomagnetic data obtained from such cores consist of a set of inclination values of known sequence but unknown spacing in time. The objective in analyzing these data is to determine the paleolatitude of the sampling site. If paleomagnetic data are available from widely separated sites on the same plate and if both the paleolatitude and its associated confidence limits are known for each site, the data may be combined to yield a paleopole for the plate and an ellipse of 95% confidence [*Gordon and Cox*, 1980]. The coordinates of the paleopole and the orientation, size, and ellipticity of the confidence ellipse depend not only on the site coordinates and paleocolatitudes but also on the magnitude of the confidence intervals. Therefore the need has arisen for a method of analysis that yields an accurate estimate of confidence limits for paleolatitude data from azimuthally unoriented vertical cores. This same method may be used to estimate paleolatitudes from paleomagnetic data obtained from crustal blocks that have rotated by varying amounts about vertical axes.

This report is concerned with several questions that emerge in the course of analyzing paleoinclination data: (1) is it better to average inclination values directly, or to convert to paleolatitudes first and then average; (2) is the mean paleolatitude an unbiased measure of the true ancient latitude at the sample site; (3) do the strata sampled in a given study span enough time to permit removal of the effects of geomagnetic secular variation by averaging; and (4) what statistical methods are appropriate for assessing confidence limits?

To summarize the results presented below, paleolatitudes are found to be preferable over paleoinclinations for analysis, but both mean paleoinclinations and mean paleolatitudes are found to be biased estimators. To convert a mean paleolatitude determined paleomagnetically to an unbiased estimate of the true paleolatitude, it is necessary to subtract a paleolatitude correction, the magnitude of which increases with increasing latitude. The variance of the paleolatitude estimate also depends on the latitude of the site. Tables are presented giving paleolatitude corrections and variances as a function of paleolatitude. This analysis is based upon assumptions about the nature of geomagnetic secular variations in the past and about the nature of the probability distribution function for paleolatitudes generated as a result of secular variation. A comparison of the distribution function used in the present analysis with paleomagnetic data from Quaternary basalts confirms that the function fits the angular dispersion of paleocolatitudes generated as a result of geomagnetic secular variation. In the final section of the report, the present method of analysis is applied to paleomagnetic data from all Deep Sea Drilling Project (DSDP) sites from the Pacific plate. The data from Cretaceous sites are then combined to obtain best fit poles for the Early and Late Cretaceous epochs.

VIRTUAL GEOMAGNETIC COLATITUDES

Paleomagnetic data from completely oriented samples may be analyzed using either field directions or virtual geomagnetic poles (VGP's). By definition, a VGP describes an instantaneous direction of the field at some sample site by specifying the colatitude Θ and longitude ϕ of the negative pole of a geocentric dipole that would produce the observed declination and inclination at the sample locality. By convention the present field with a negative pole in the northern hemisphere is of normal polarity. The use of VGP's commonly turns out to be simpler than the use of field directions because of the nature of the symmetry of the angular dispersion produced by secular variation (SV). A major part of the SV is pro-

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Paper number 3R1307.
0034-6853/84/003R-1307$15.00

duced by the symmetrical wobble of a geocentric dipole about the earth's rotation axis. If the resulting dipolar angular dispersion is contoured for density, the contours of field directions at the equator are quasi-ellipses twice as long in the vertical as in the horizontal direction. Because of this asymmetry, the field directions cannot be analyzed by simple one-parameter statistical models like that of *Fisher* [1953]. Similarly, at the equator the present non dipole field produces about twice as much angular dispersion in the vertical as in the horizontal direction [*Cox*, 1959]. If a set of magnetic vectors representing SV produced by both dipole and non dipole contributions is transformed to a set of VGP's using the dipole equations, the distribution of VGP's about their mean will be symmetrical and can be analyzed using one parameter statistical models. VGP's, in addition to having more nearly symmetrical distributions than field directions, also have angular dispersions that are less latitude dependent than those of field directions [*Cox*, 1970]. The angular standard deviation of VGP's due to SV decreases from a value of about 20° at the poles to about 13° at the equator, as is discussed more fully below.

Because of their greater symmetry and smaller variation with latitude, VGP's are commonly used in the analysis of paleomagnetic data for which both declination and inclination are known. However, in analyzing the inclination data from azimuthally unoriented vertical cores, VGP's cannot be calculated because the declinations are not known. For analyzing data of this type the concept of virtual geomagnetic latitude (VGL) introduced by *Grommé and Vine* [1972] is especially useful. If I_i is the inclination from one volcanic cooling unit or other stratum corresponding to a nearly instantaneous reading of the paleofield, then the virtual geomagnetic latitudes (VGL's) λ_i and colatitudes (VGC's) Θ_i are calculated from the inclination values by

$$\lambda_i = \tan^{-1}\left[\frac{1}{2}\tan I_i\right] \quad (-90° \leqslant \lambda_i \leqslant 90°) \qquad (1a)$$

$$\Theta_i = \cot^{-1}\left[\frac{1}{2}\tan I_i\right] \quad (0° \leqslant \Theta_i \leqslant 180°)$$

$$= 90° - \lambda_i \qquad (1b)$$

These equations lead to the result that at the equator the standard deviation of paleolatitudes is less than that of inclinations whereas at the poles it is greater [*Cox*, 1970]. If symmetrical confidence limits are found for a mean inclination, the corresponding colatitude confidence limits found by transforming the inclination limits using (1*b*) will be distributed asymmetrically about the mean colatitude. This asymmetry is reduced if the inclination I_i for each stratum is first transformed to a VGC value Θ_i and the Θ_i's are then averaged.

Probability Distribution Function for Virtual Colatitudes

We begin with the usual assumption that VGP's calculated from paleomagnetically determined paleofield directions are distributed with azimuthal symmetry about a pole P (Figure 1) and that the probability distribution function (PDF) for angular scatter depends only on the polar angle Θ':

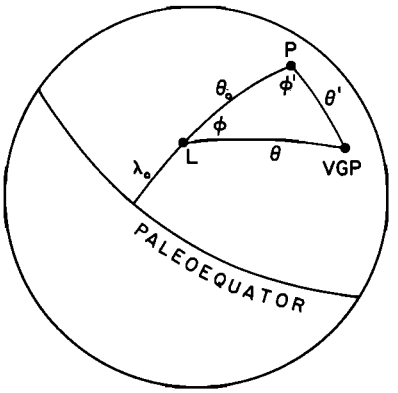


Fig. 1. VGP is a pole from a population of virtual geomagnetic poles dispersed with azimuthal symmetry about the mean pole P. The probability density function of VGP's about P depends upon Θ' but not upon ϕ'. At drill site L the virtual geomagnetic colatitude Θ determined in a sampled stratum is the distance from L to the VGP for that stratum, whereas the true paleocolatitude Θ_0 is the distance from L to the mean pole P. The probability density function for colatitudes Θ is shown in Figure 2.

$$F(\Theta', \phi')dA = \frac{K}{4\pi \sinh K} \exp(K \cos \Theta')dA \qquad (2)$$

where $F(\Theta', \phi')\,dA$ is the probability of a VGP within a small area dA on the unit sphere displaced an angle Θ' from the mean VGP [*Fisher*, 1953]. The basic data to be analyzed consist of a set of values of Θ_i (VGC's) found by (1) from a set of instantaneous values of I_i determined paleomagnetically from i strata drilled at site L. At the outset it should be noted that the mean value $\bar{\Theta}$ of Θ_i yields a biased estimate of the true paleocolatitude, Θ_0. At $\Theta_0 = 0$, for example, $\Theta_i \geqslant \Theta_0$ and in general $\bar{\Theta} > \Theta_0$. Substantial statistical errors are introduced, especially at high latitudes, if Θ is treated as a normally distributed random variable and if $\bar{\Theta}$ is used as an unbiased estimate of the true paleocolatitude.

Our analysis of colatitudes follows that of *Briden and Ward* [1966] for analyzing inclinations from borehole data for which declinations are not known. The key to the analysis is finding the PDF of colatitudes Θ, where Θ is the angular distance from the sampling site L to the VGP (Figure 1). Since the original PDF is symmetrical about the true pole P, the probability $F(\Theta')dA$ that a VGP will lie within a small, solid angle dA at an angular distance Θ' from P is independent of the angle ϕ'. However, the PDF $F(\Theta, \phi)$ of VGP's about L depends on the angle ϕ. Noting that

$$\cos \Theta' = \cos \Theta_0 \cos \Theta + \sin \Theta_0 \sin \Theta \cos \phi \qquad (3)$$

the PDF may be expressed in coordinates (Θ, ϕ) about L:

$$F(\Theta, \phi) = \frac{K}{4\pi \sinh K} \exp[K(\cos \Theta_0 \cos \Theta + \sin \Theta_0 \sin \Theta \cos \phi)] \qquad (4)$$

Since values of Θ but not of ϕ are known from our paleomagnetic data, the marginal distribution giving the PDF of

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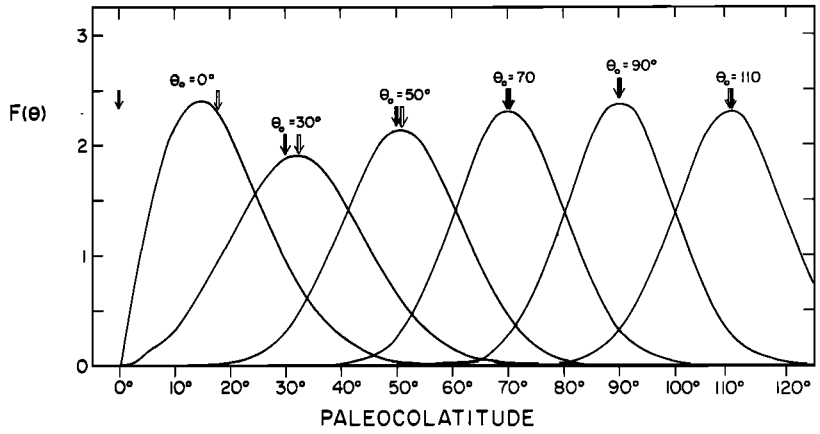


Fig. 2. Probability density function, $F(\Theta)$, for paleocolatitudes. The parameters of the PDF (equation (5)) are the true colatitude, Θ_0, of the sampling site and Fisher's precision parameter, K_{SV}, for geomagnetic secular variation at the colatitude of the site (Table 2). Solid Arrow indicates true paleocolatitude Θ_0. Open Arrow indicates expected mean paleocolatitude for random sample of measurements at a site with true paleocolatitude Θ_0; the expected mean paleocolatitude is the expectation of the PDF. For $\Theta_0 = 0$ the true paleocolatitude (solid arrow) and expected mean paleocolatitude (open arrow) differ by 18°.

Θ independent of ϕ is needed. Integrating with respect to ϕ around a ring of radius Θ,

$$F(\Theta) = \int_0^{2\pi} \frac{K}{4\pi \sinh K} \sin\Theta \exp\left[K(\cos\Theta_0 \cos\Theta + \sin\Theta_0 \sin\Theta \cos\phi)\right] d\phi \quad (5)$$

where $F(\Theta)d\Theta$ is the probability that an observed VGC will have a value between Θ and $\Theta + d\Theta$. This equation has the same form as that for the PDF of borehole inclination values derived by *Briden and Ward* [1966]. Noting that the integral form of the zero-order hyperbolic Bessel function is

$$I_0(x) = \frac{1}{2\pi} \int_0^{2\pi} \exp\left[x \cos\phi\right] d\phi \quad (6)$$

(5) becomes

$$F(\Theta) = \frac{K}{2 \sinh K} \sin\Theta \exp\left[K \cos\Theta_0 \cos\Theta\right] \times I_0\left(K \sin\Theta_0 \sin\Theta\right) \quad (7)$$

The two parameters of this PDF are the true paleocolatitude, Θ_0, of the sample site and the precision parameter, K, describing the angular dispersion of VGP's. Note (Figure 2) that if a sampling site is located at the paleoequator ($\Theta_0 = 90°$), then the distribution function $F(\Theta)$ is symmetrical, the maximum occurs at $\Theta = \Theta_0$, and the mean or expectation of the distribution is equal to Θ_0. However, if the sampling site is not at the paleoequator ($\Theta_0 \neq 90°$), then the distribution function is not symmetrical, the maximum does not occur at the true paleocolatitude of the site, and the mean of the distribution is not equal to the true paleocolatitude. Because of this asymmetry, the mean of a set of paleocolatitude data does not provide an unbiased estimate of the true paleocolatitude Θ_0. Before considering how the bias can be removed, we will discuss some important differences between the bivariate Fisher distribution of (2) and the univariate distribution of (7).

Univariate Versus Bivariate Distributions

In deriving (7) for $F(\Theta)$ from (2) for $F(\Theta', \phi')$, we began with $F(\Theta', \phi')$, Fisher's bivariate or two-dimensional probability density per unit area defined over the surface of a unit sphere, and from this derived by integration the univariate or one-dimensional distribution $F(\Theta)$, which varies over the range $0° \leqslant \Theta \leqslant 180°$. The bivariate and univariate distributions have different maxima, means, and angular dispersions. Although two-dimensional distributions have sometimes been used in the literature where one-dimensional distributions were appropriate and vice versa, the two cannot be used interchangeably in establishing confidence limits.

For example, *Grommé and Vine* [1972] assumed that the error of the paleocolatitude of Midway Atoll was equal to the radius of confidence circle of the Hawaiian seamounts, and similarly *Kono* [1980] assumed that the confidence limits on colatitudes of various Emperor seamounts were equal to the confidence circles about the best fit poles. In both cases, bivariate confidence limits were used inappropriately. In general, univariate distributions are appropriate for testing the hypothesis that the pole corresponding to an observed set of paleocolatitudes lies at a particular distance from a site, whereas bivariate distributions are appropriate for testing hypotheses about the location of poles on the globe. An analogous argument applies to the analysis of paleomagnetic inclinations and declinations [*Demarest*, 1983].

Figure 3 illustrates why bivariate and univariate confidence limits are not interchangeable. The shaded circle around the hypothetical paleomagnetic pole P describes the angular standard error of the pole which, by definition, contains 63% of the probability distribution for the pole. A hypothetical drill site, L, at the center of the figure has a paleocolatitude, Θ, where Θ is the angular distance between the pole and the drill site. If the erroneous assumption is made that the standard error of the paleocolatitude

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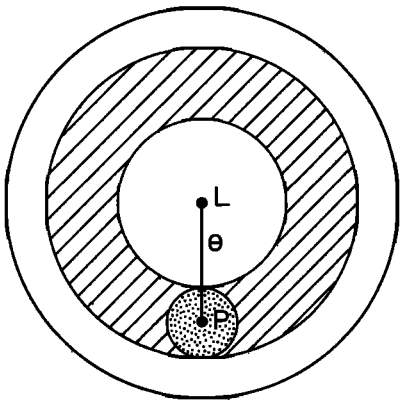


Fig. 3. Example demonstrating why univariate and bivariate confidence intervals are not interchangeable. At drill site L a paleocolatitude Θ is determined with an angular standard error s_u. At pole P the shaded circle has radius s_b, the standard error of the pole, and by definition contains 63% of the probability distribution of the pole. If the erroneous assumption is made that $s_u = s_b$, the corresponding confidence limit for Θ is the annular region shown by diagonal shading. Since the fraction of the total probability of P lying within the annular region is greater that 0.63, setting $s_u = s_b$ clearly gives too large a value of s_u.

Θ is equal to the standard error of the pole, the corresponding confidence interval for Θ is the annular region shown by diagonal shading. The fraction of the total probability of P lying within this annular region is clearly more than 0.63 because that much is included within the shaded circle about P and all of that circle lies within the annular region. In fact, if $\Theta = 90°$, then it can be shown that 84% of the probability for the pole lies within the annular region. This example illustrates the general relationship discussed later that univariate confidence limits for paleocolatitudes are smaller than the bivariate limits for poles.

Some useful analogs are provided by the familiar case of univariate and bivariate Gaussian distributions on a plane. The planar analog of Fisher's bivariate distribution $F(\Theta', \phi')$ is the bivariate Gaussian distribution,

$$P(x, y)\, dA = \frac{1}{2\pi s^2} \exp\left[\frac{-(x^2 + y^2)}{2s^2}\right] dA \qquad (8)$$

where x and y are random, zero mean variables with variance s^2 and $dA = dx\, dy$ an element of area. This distribution can arise from the joint probability density of two uncorrelated univariate Gaussian distributions,

$$P(x, y)\, dx\, dy = P(x)P(y)\, dx\, dy \qquad (9)$$

where

$$P(x)\, dx = \frac{1}{(2\pi)^{1/2} s_x} \exp\left[\frac{-x^2}{2{s_x}^2}\right] dx \qquad (10a)$$

and

$$P(y)\, dy = \frac{1}{(2\pi)^{1/2} s_y} \exp\left[\frac{-y^2}{2{s_y}^2}\right] dy \qquad (10b)$$

for which we assume that $s = s_x = s_y$. This condition establishes the azimuthal symmetry of $P(x, y)$, permitting simplification with a change of variables to radial distance $r = (x^2 + y^2)^{1/2}$ and polar angle ϕ. Noting that $dA = dx\, dy = r\, dr\, d\phi$,

$$P(r, \phi)\, dA = \frac{1}{2\pi s^2} \exp\left[\frac{-r^2}{2s^2}\right] dA \qquad (11)$$

This two-dimensional distribution has a maximum density per unit area at the origin and is the planar analog of the two-dimensional Fisher distribution. The second moment of the angular distance r,

$$E(r^2) = \int_0^{2\pi} \int_0^{\infty} r^2\, P(r, \phi)\, r\, dr\, d\phi = 2s^2 \qquad (12)$$

is twice as large as the variance s^2 of the univariate distributions $P(x)$ and $P(y)$.

A one-dimensional distribution in the single variable r can be obtained from (11) by integrating around a ring of constant r,

$$P(r)\, dr = \frac{r}{s^2} \exp\left[\frac{-r^2}{2s^2}\right] dr \qquad (13)$$

Unlike the two-dimensional distribution, this function has a value of zero at the origin ($r = 0$) and a maximum at $r = s$. The first and second moments of this function are

$$E(r) = \int_0^{\infty} r\, P(r)\, dr = (\pi/2)^{1/2}\, s \qquad (14)$$

$$E(r^2) = \int_0^{\infty} r^2\, P(r)\, dr = 2s^2 \qquad (15)$$

and the variance is

$$E(r^2) - E[(r - \bar{r})^2] = E(r^2) - E^2(r)$$

$$= s^2\left[2 - \frac{\pi}{2}\right] = 0.43s^2 = (0.66s)^2 \qquad (16)$$

where, as before, s^2 is the variance of the univariate distributions $P(x)$ and $P(y)$ from which the bivariate distribution was derived.

The following is the planar analog of finding a paleomagnetic pole using paleocolatitude data from two sites located on the equator 90° apart (Figure 4). At two sites U and V

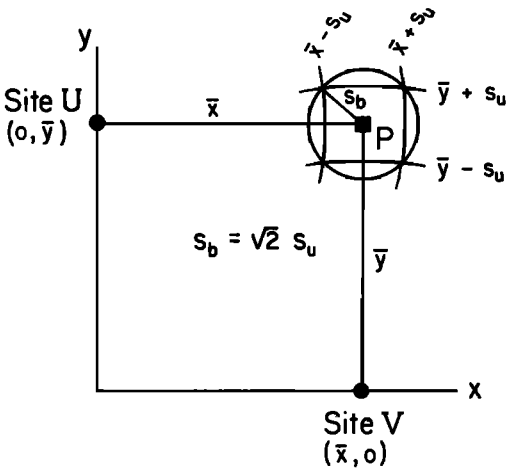


Fig. 4. Planar analog of finding a pole P from two sites, U and V, on a sphere for the simple case where arcs UP and VP are perpendicular. Assume that experimental results show that the distances (colatitudes) from sites U and V to P are $\bar{x} \pm s_u$ and $\bar{y} \pm s_u$, respectively, where s_u is the univariate standard error of both $\bar{x}$ and $\bar{y}$. Then the bivariate standard error, s_b, in the location of P is $\sqrt{2}s_u$.

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with coordinates $(0, \bar{y})$ and $(\bar{x}, 0)$ respectively, the distances to the pole are random variables x and y, the means of which are known from observations to be $\bar{x}$ and $\bar{y}$ and the standard errors of which are s. The two means are the analogs of the two paleocolatitudes. If $\bar{x}$ and $\bar{y}$ are both much larger than s, then the distribution function for the pole is approximated by the bivariate probability $P(x, y) = P(x)P(y)$, where $P(x)$ and $P(y)$ are univariate Gaussian distributions with means $\bar{x}$ and $\bar{y}$ and variances s^2. The best estimate of the location of the pole is $(\bar{x}, \bar{y})$ and the corresponding variance for $P(x, y)$ is $2s^2$, as discussed earlier. Thus if s_u is the standard error of the means of the two sets of univariate data, then the corresponding bivariate standard error s_b for the pole is

$$s_b = (2)^{1/2} s_u \tag{17}$$

The standard error for the pole will be a circle about $(\bar{x}, \bar{y})$ with a radius of $(2)^{1/2}s_u$.

The following is the planar analog of a paleomagnetic pole located at or near the sampling site ($\Theta_0 \approx 0$). In this case, the radial distance r is the analog of the colatitude Θ. Observations of the single variable r are described by the univariate distribution $P(r)$. Specifying the location of the pole is a two-dimensional problem for which the appropriate variance is that of the bivariate distribution $P(r, \phi)$. By (12) and (16) the ratio of the two variances is

$$\frac{E(r^2)}{E[(r - \bar{r})^2]} = \frac{2}{2 - \pi/2} \tag{18}$$

If s_u is the standard error of the observed mean value of r, then the corresponding bivariate standard error s_b for the pole will be

$$s_b = \frac{(2)^{1/2}}{(2 - \pi/2)^{1/2}} s_u = 2.16\, s_u \tag{19}$$

Turning now to the sphere, the analogy between the Gaussian and Fisherian distributions may be seen by making the small angle approximation $\cos \Theta' \approx 1 - (\Theta')^2/2$ in (2), which gives

$$F(\Theta', \phi') \approx \frac{K}{4\pi \sinh K} \exp\left[\frac{K(1 - (\Theta')^2)}{2}\right] \approx \frac{K}{2\pi} \exp\left[\frac{-K(\Theta')^2}{2}\right] \tag{20}$$

The angular variance S_b^2 of this bivariate distribution is

$$S_b^2 = E[(\Theta')^2] = 2/K \tag{21}$$

For small angles Θ', the bivariate Fisherian distribution is approximated by the bivariate Gaussian distribution upon replacement of Θ' by r and K by $1/S^2$.

The univariate density $F(\Theta')$ of colatitudes drawn from a Fisherian distribution, which is the analog of the univariate planar distribution $P(r)$, is found by integrating around a circle of constant Θ':

$$F(\Theta') = \int_0^{2\pi} F(\Theta', \phi') \sin \Theta'\, d\phi' \approx K\Theta' \exp\left[\frac{-K(\Theta')^2}{2}\right] \tag{22}$$

(Note that $F(\Theta')$ is simply a function of K, whereas $F(\Theta)$ as defined by (7) is a function of both Θ_0 and K.) The first and second moments of $F(\Theta')$ are

$$E(\Theta') = \int_0^{\pi} \Theta' F(\Theta')\, d\Theta' \approx (\pi/2K)^{1/2} \tag{23}$$

$$E[(\Theta')^2] = \int_0^{\pi} (\Theta')^2 F(\Theta')\, d\Theta' \approx 2/K \tag{24}$$

and the variance S_u^2 is

$$S_u^2 = E[(\Theta')^2] - E^2[\Theta'] = \frac{2 - \pi/2}{K} = \frac{0.43}{K} \tag{25}$$

For small angles Θ', the univariate distribution $F(\Theta')$ is approximated by the univariate circular Gaussian distribution $P(r)$ if Θ' and K are replaced by r and $1/s^2$. The approximations are valid within several percent for values of $K \geqslant 12.5$ [*Cox*, 1970].

If a set of VGP's is distributed with bivariate angular variance $S_b^2 = 2/K$, then for a site located at the mean pole ($\Theta_0 = 0$ and $\Theta' = \Theta$) the variance for the corresponding set of colatitudes will be $S_u^2 = (2 - \pi/2)/K$, so that $S_b^2/S_u^2 = 2/(2 - \pi/2)$. If S_u is the angular deviation of a set of paleocolatitude data, then the corresponding standard deviation for the VGP's will be

$$S_b = S_u \frac{(2)^{1/2}}{(2 - \pi/2)^{1/2}} = S_u \frac{(2)^{1/2}}{0.66} = 2.16 S_u \tag{26}$$

At all colatitudes the bivariate angular variances for poles are greater than the corresponding angular variances for paleocolatitudes by a factor of at least 2.

The ratio of the confidence limits for the bivariate and univariate data is not simply S_b/S_u because the two distribution functions are different. For a bivariate Fisherian distribution of VGP's with precision parameter K and angular variance $S_b^2 = 2/K\,(\text{rad}^2) = 81^2/K\,(\text{deg}^2)$, the 95% confidence limit for the mean pole is a circle with radius α_{95}, where $\alpha_{95} = 140/(KN)^{1/2}$. Therefore the 95% confidence limit is related to the angular variance of the population, S_b^2, by $\alpha_{95} = 1.73\, S_b/(N)^{1/2}$. For the corresponding univariate distribution of colatitudes with variance S_u^2, the usual assumption is that the distribution of colatitudes is Gaussian and that the 95% confidence interval, β_{95}, for the mean colatitude is $\beta_{95} = 1.96 S_u/(N)^{1/2}$ where $S_u/(N)^{1/2}$ is the standard error of the mean. As noted earlier (Figure 2), the assumption that $F(\Theta)$ is Gaussian is more nearly valid at the equator ($\Theta_0 = 90°$) than at the pole ($\Theta_0 = 0°$). Under the assumption of a Gaussian distribution, $\alpha_{95}/\beta_{95} = 0.88(S_b/S_u)$. For colatitudes $\Theta_0 > 30°$ we will show that $(S_b/S_u) \approx (2)^{1/2}$. Thus, the ratio of bivariate to univariate 95% confidence intervals is

$$\frac{\alpha_{95}}{\beta_{95}} = 0.88(2)^{1/2} = 1.25 \quad (\Theta_0 > 30°) \tag{27}$$

At the pole ($\Theta_0 = 0°$) where $(S_b/S_u) = 2.16$, $(\alpha_{95}/\beta_{95}) = 1.9$. Thus the radius of the bivariate 95% confidence circle is larger than the half-length of the univariate 95% confidence interval by a factor that varies from 1.25 to 1.9. Given N colatitudes with angular standard deviation S_u, confidence limits for the corresponding set of poles can be estimated as

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TABLE 1. Angular Standard Deviation σ_u of Colatitudes Produced at Colatitude Θ_0 by the Angular Dispersion of VGP's With Standard Deviations σ_b

	σ_b												
Θ_0	2.5	5.0	7.5	10.0	12.5	15.0	17.5	20.0	22.5	25.0	30.0	35.0	40.0
0	1.2	2.3	3.5	4.7	5.8	7.0	8.2	9.5	10.7	12.0	14.6	17.4	20.4
2	1.4	2.5	3.6	4.8	5.9	7.1	8.3	9.5	10.7	12.0	14.6	17.4	20.4
5	1.7	3.0	4.1	5.1	6.2	7.4	8.5	9.7	10.9	12.2	14.8	17.5	20.5
10	1.8	3.4	4.8	6.0	7.1	8.2	9.2	10.4	11.5	12.7	15.2	17.9	20.8
15	1.8	3.5	5.1	6.5	7.8	9.0	10.1	11.2	12.3	13.5	15.9	18.5	21.2
20	1.8	3.5	5.2	6.8	8.3	9.6	10.8	12.0	13.2	14.3	16.6	19.2	21.8
25	1.8	3.5	5.2	6.9	8.5	10.0	11.4	12.7	13.9	15.1	17.4	19.9	22.5
30	1.8	3.5	5.3	7.0	8.6	10.2	11.7	13.2	14.5	15.7	18.2	20.6	23.2
40	1.8	3.5	5.3	7.0	8.8	10.4	12.1	13.7	15.2	16.7	19.4	22.0	24.5
50	1.8	3.5	5.3	7.0	8.8	10.5	12.2	13.9	15.6	17.2	20.2	23.0	25.6
60	1.8	3.5	5.3	7.1	8.8	10.6	12.3	14.0	15.8	17.4	20.7	23.7	26.4
70	1.8	3.5	5.3	7.1	8.8	10.6	12.4	14.1	15.8	17.6	20.9	24.1	27.0
80	1.8	3.5	5.3	7.1	8.8	10.6	12.4	14.1	15.9	17.6	21.1	24.3	27.3
90	1.8	3.5	5.3	7.1	8.8	10.6	12.4	14.1	15.9	17.6	21.1	24.4	27.4

$$\alpha_{95} \approx 0.88\left[\frac{S_b}{S_u}\right] \beta_{95} = 1.72\left[\frac{S_b}{S_u}\right]\frac{S_u}{(N)^{1/2}} \qquad (28)$$

where S_b/S_u can be evaluated using Table 1.

Turning to the general problem of how errors are propagated when poles are determined from colatitudes measured at widely separated sites, consider the simple case of two deep sea drilling sites located on the equator at 0° east and 90° east longitude. At each site a mean colatitude of $90° \pm s$ is determined, where s is the angular standard error of the mean. The best fit pole obviously lies at the pole, 90°N, which is at the intersection of the great circles centered on the two sites. The propagation of errors can best be traced using a plane approximation to represent the surface of the sphere in the vicinity of the pole. Let the center of planar coordinates be at the pole with the x and y axes directed along the 0°E and 90°E longitude circles. If the univariate distributions for the two sets of colatitudes are described by PDF's with standard deviations of s, then the corresponding distributions $P(x)\,dx$ and $P(y)\,dy$ for the pole have means of zero and the same standard deviations and the joint bivariate distribution for the pole is $P(x, y)\,dx\,dy = P(x)P(y)\,dx\,dy$. As the errors in x and y are uncorrelated, the covariance matrix for the joint PDF for the pole is

$$\begin{bmatrix} s_{xx}^2 = s^2 & s_{xy}^2 = 0 \\ s_{yx}^2 = 0 & s_{yy}^2 = s^2 \end{bmatrix} \qquad (29)$$

where

$$s_{ij}^2 = \int\int (u_i - \bar{u}_i)(u_j - \bar{u}_j) P(u_i, u_j)\,du_i\,du_j \qquad (30)$$

Because in this example $s_{xx}^2 = s_{yy}^2$, contours of equal probability are circles. In the general case where $s_{xx}^2 \neq s_{yy}^2$ the contours are ellipses. In the circular case letting $r^2 = x^2 + y^2$, the variance $E(r^2) = E(x^2) + E(y^2)$ was previously shown to equal $2s^2$. Thus the appropriate radius for the standard error circle about the pole is $(2)^{1/2}$ times the standard error s of the colatitude measurements. Similarly when $s_{xx} \neq s_{yy}$ and elliptical confidence intervals are appropriate, the lengths of the semiaxes of the standard error ellipses are given by $(2)^{1/2}s_{xx}$ and $(2)^{1/2}s_{yy}$. In the general case when the covariance matrix has been determined from standard errors of data from many sites, the covariance matrix is not initially diagonal ($s_{xy} \neq 0$) but can always be diagonalized by a change of coordinates, after which the above reasoning applies.

To examine more precisely the relationship on a sphere between the bivariate angular dispersion of poles and the univariate angular dispersion of colatitudes, the expression for the variance of colatitudes,

$$\sigma_u^2 = E(\Theta^2) = \int_0^{\pi} [\Theta - E(\Theta)]^2 F(\Theta)\,d\Theta \qquad (31)$$

was evaluated by numerical integration (Table 1) for a wide range of pole dispersions, σ_b. These results show, for example, that VGP's with an angular dispersion of $\sigma_b = 20°$ (Table 1) would produce colatitudes with an angular standard deviation of $\sigma_u = 9.5°$ at the pole ($\Theta_0 = 0$) and of $\sigma_u = 14.1°$ at the equator ($\Theta_0 = 90°$), in accord with the earlier generalization that $s_b/s_u \geq (2)^{1/2}$. A plot of the ratio $(\sigma_b/(2)^{1/2}\sigma_u)$ (Figure 5) shows that except at high latitudes ($\Theta_0 < 35°$), the relationship $\sigma_b \approx (2)^{1/2}\sigma_u$ is valid to within 5% for pole dispersions no greater than $\sigma_b = 20°$.

Angular Dispersion Caused by SV

Two options are open for evaluating the paleosecular variation precision parameter, K_{SV}. The first is to evaluate K_{SV} from the variance of the paleocolatitudes observed at a particular sampling site. The second option is to use a global model for K_{SV} based on prior studies of paleosecular variation. Whether the first option is viable at a given site depends upon the extent of the sampling. For bivariate data, the number N of strata needed to determine $K_{SV} \pm 0.25K_{SV}$ at 95% confidence is $N = 50$ [*Cox*, 1969]. Moreover, to measure K_{SV} with even 25% accuracy from 50 strata, the times of formation of the individual strata must be separated by the longest period present in SV, which is thought to be at least 10^4 years. This condition is rarely fulfilled by the sequence of lava flows encountered in a single drill core. Commonly, the angular dispersion of magnetic directions from some particular sequence of strata is observed to be anomalously small, not because SV was unusually small during the period in which the strata formed, but rather because the time of formation was so brief that the full range of SV was not sampled. Under the difficult conditions of drilling deep-sea basalts it may

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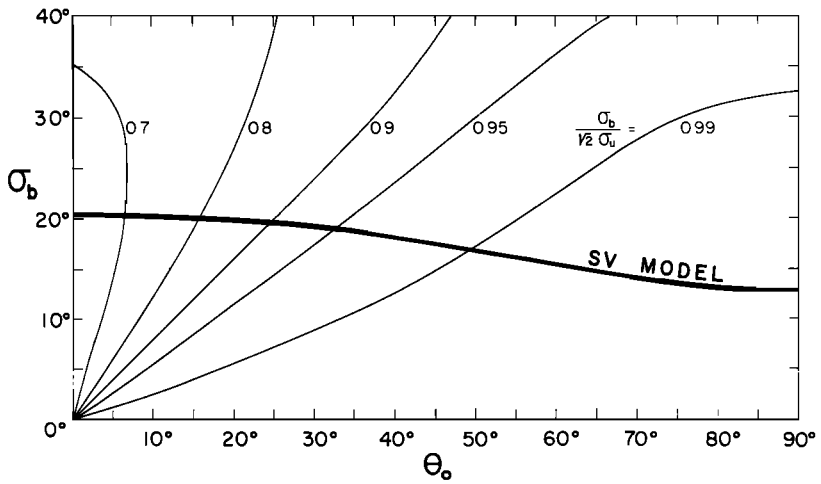


Fig. 5. Curves for finding the univariate angular standard deviation of colatitudes, σ_u, given the bivariate angular standard deviation of VGP's, σ_b, at a site with colatitude Θ_0. Along the 0.99 contour, for example, $\sigma_u = \sigma_b / 0.99\sqrt{2} = 0.714\ \sigma_b$. The heavy curve shows the angular standard deviation, σ_{SV}, of our model of secular variation as a function of site colatitude, Θ_0. For models of secular variation with angular dispersion no greater than ours, at all colatitudes $\Theta > 30°$ the approximation $\sigma_b \approx \sqrt{2}\sigma_u$ is accurate within 5%.

rarely prove possible to penetrate enough strata to fully sample geomagnetic SV.

The second option is to use a value of K_{SV} found by evaluating a much larger set of paleomagnetic data from sites with similar paleocolatitudes. A marked increase in the dispersion of VGP's from low values at the equator to high values at the poles is evident in global analyses of paleomagnetic data. To describe this increase, several models that incorporate the effects of dipole wobble, dipole intensity fluctuations, variations with latitude of the amplitude of the non dipole field, and drift of the non dipole field have been advanced [*Cox*, 1962, 1970; *Creer*, 1962; *Irving and Ward*, 1964; *McElhinny and Merrill*, 1975; *Harrison*, 1980]. Although differing in details, these models all provide useful descriptions of the observed variation with latitude of paleosecular variation. Calculations for the present analysis were made using values of K_{SV} (Table 2) corresponding to *Harrison*'s [1980] model for the latitude dependence of SV, which fits the observed latitude dependence of angular dispersion of paleomagnetic data for the past 5 million years [*McElhinny and Merrill*, 1975]. If, as is quite possible, the amount of SV has varied with longitude as well as latitude or if it has been systematically greater or smaller in the past, the paleolatitude corrections and confidence limits derived in the present analysis will require modification. However, until values of SV as a function of latitude are determined reliably for different regions and different geologic periods, the values of K_{SV} listed in Table 2 provide a reasonable basis for estimating paleocolatitude corrections and confidence limits.

The model for secular variation discussed above and described in Table 2 yields the probability density functions for colatitude shown in Figure 2. The PDF is symmetrical at the equator and becomes highly asymmetrical only when the sample site is close to the poles. The paleocolatitude corrections, which correspond to the separations between open and solid arrows in Figure 2, are also greatest near the poles.

Paleocolatitude Correction

The next step of our analysis is to find the appropriate paleocolatitude correction T such that $(\bar{\Theta} - T)$ provides an unbiased estimate of the true paleocolatitude Θ_0, i.e.,

$$\Theta_0 \approx \bar{\Theta} - T \tag{32}$$

The following approximate method is suggested for evaluating $\bar{\Theta}$ and T. The mean is first found from a data set of N observations by

$$\bar{\Theta} = \frac{1}{N}\sum_{i=1}^{N}\Theta_i \tag{33}$$

and $\bar{\Theta}$ is then taken as an estimate of the expectation $E(\Theta)$, where

$$E(\Theta) = \int_0^{\pi} \Theta\, F(\Theta)\, d\Theta \tag{34}$$

Since $F(\Theta)$ is a function of both K_{SV} and Θ_0, so is $E(\Theta)$. Values of the expectation $E(\Theta)$ were found by numerical integration of (34) for values of K_{SV} from 4 to 1000, which spans the range of angular dispersion commonly encountered in paleomagnetism. In presenting the results of these integrations in Table 3, the quantity $\sigma_{SV} = 81°/(K_{SV})^{1/2}$ is listed rather than K_{SV} in order to bring out relationships between σ_{SV} and $E(\Theta)$. Note that for small values of Θ_0, $E(\Theta) \approx \sigma_{SV}$. Note also the symmetry of $E(\Theta)$ about the equator: For a value of $\sigma_{SV} = 20°$, the expectations for $\Theta_0 = 60°$ and $\Theta_0 = 120°$ are both dis-

TABLE 2. Model for Paleosecular Variation Based on Late Cenozoic Geomagnetic Field

	Paleocolatitude Θ_0, deg									
	0	10	20	30	40	50	60	70	80	90
Precision parameter K_{SV}	15.8	16.0	16.7	18.0	20.0	23.2	27.7	33.3	38.4	40.5
$\sigma_{SV} = 81°/\sqrt{K_{SV}}$	20.4	20.3	19.8	19.1	18.1	16.8	15.4	14.0	13.1	12.7

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TABLE 3. Expectation of Colatitude, $E(\Theta)$, as a Function of Angular Secular Variation, σ_{SV}, and Paleocolatitude, Θ_0

	$\sigma_{SV} = 81°/\sqrt{K_{SV}}$												
Θ_0	2.5	5.0	7.5	10.0	12.5	15.0	17.5	20.0	22.5	25.0	30.0	35.0	40.0
0	2.2	4.4	6.7	8.9	11.1	13.4	15.6	17.9	20.1	22.4	27.1	31.8	36.7
2	2.9	4.8	6.9	8.1	11.3	13.5	15.7	18.0	20.2	22.5	27.1	31.9	36.8
5	5.3	6.4	8.1	10.0	12.0	14.1	16.2	18.4	20.6	22.9	27.4	32.1	37.0
10	10.2	10.6	11.5	12.8	14.4	16.1	18.0	20.0	22.0	24.1	28.5	33.0	37.7
15	15.1	15.4	16.0	16.8	17.9	19.2	20.7	22.4	24.2	26.1	30.1	34.4	38.9
20	20.1	20.3	20.7	21.3	22.0	23.0	24.2	25.6	27.1	28.8	32.4	36.4	40.6
25	25.1	25.2	25.5	26.0	26.5	27.3	28.2	29.3	30.6	32.0	35.2	38.7	42.7
30	30.0	30.2	30.4	30.8	31.2	31.8	32.5	33.4	34.4	35.6	38.3	41.5	45.1
40	40.0	40.1	40.3	40.5	40.8	41.2	41.7	42.3	42.9	43.7	45.7	48.1	51.0
50	50.0	50.1	50.2	50.4	50.6	50.9	51.2	51.6	52.0	52.6	53.9	55.7	57.8
60	60.0	60.1	60.1	60.3	60.4	60.6	60.8	61.1	61.4	61.7	62.6	63.8	65.4
70	70.0	70.0	70.1	70.2	70.3	70.4	70.5	70.7	70.9	71.1	71.6	72.4	73.4
80	80.0	80.0	80.0	80.1	80.1	80.2	80.2	80.3	80.4	80.5	80.8	81.1	81.6
90	90.0	90.0	90.0	90.0	90.0	90.0	90.0	90.0	90.0	90.0	90.0	90.0	90.0
100	100.0	100.0	100.0	99.9	99.9	99.8	99.8	99.7	99.6	99.5	99.2	98.9	98.4

placed 1.1° toward the equator from 60° and 120°. The colatitude correction

$$T = E(\Theta) - \Theta_0 \tag{35}$$

is plotted in Figure 6, together with the preferred model for secular variation based on paleomagnetic data for the past 5 m.y. Note that for our SV model, colatitude corrections larger than 2° are encountered only for colatitudes less than 40°.

Numerical integrations were repeated at 1° intervals of Θ_0 using values of K_{SV} obtained from (18) of *Harrison* [1980]. In (35) for the colatitude correction, integral values of Θ_0 were used as input, but the corresponding values of $E(\Theta)$ were generally not integral. Since it is the latter that are equated to the mean of an observed set of colatitudes, integral values of $E(\Theta)$ were found by interpolation to facilitate use of the table. The final result of these calculations is presented in Table 4 as $T[E(\Theta)] \approx T(\bar{\Theta})$, where the observational mean colatitude $\bar{\Theta}$ is to be used as an estimate of the theoretical value of $E(\Theta)$. The unbiased best estimate Θ^* of the colatitude is thus

$$\Theta^* = \bar{\Theta} - T(\bar{\Theta}) \tag{36}$$

Our secular variation model yields values of $E(\Theta)$ that are nowhere less than 18°, even at the pole ($\Theta_0 = 0$) (Figure 7). Therefore if the number, N, of sampled strata is large and if geomagnetic secular variation as described by the present model is fully sampled, values of $\bar{\Theta}$ less than 18° are not expected. If a value of $\bar{\Theta} < 18°$ is found from a data set, the interpretation made here is that this reflects either incomplete sampling of SV or lower values of SV than those assumed in the model. In the absence of additional information, the best estimate of the true paleocolatitude is taken to be zero. In Table 4 a linear decrease from $T(\bar{\Theta}) = 18°$ at $\bar{\Theta} = 18°$ to $T(\bar{\Theta}) = 0$ at $\bar{\Theta} = 0$ achieves this and avoids the calculation of physically meaningless negative values of Θ_0.

An alternative method of analysis is to find the mean value of $\sin \Theta_i$ and $\cos \Theta_i$ and equate these means to the theoretical expressions for the expectations of $\sin \Theta$ and $\cos \Theta$ [*Briden and Ward*, 1966]. These two equations may then be solved numerically for the two unknowns Θ_0 and K_{SV}. *Kono* [1980] employed a similar approach, using the expectations of the functions $\cos \Theta$ and $\cos^2 \Theta$. This estimate of K was shown to be unstable for small values of Θ_0 by *McFadden and Reid* [1982], who used an alternative maximum likelihood method to evaluate Θ_0 and K_{SV}. Since in the present analysis the value of K_{SV} is not calcu-

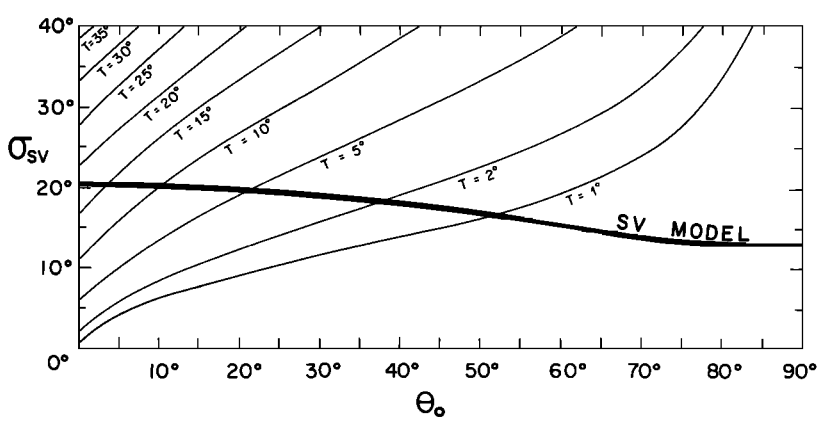


Fig. 6. The paleocolatitude correction, T, that must be subtracted from the mean paleocolatitude, $\bar{\Theta}$, to obtain an unbiased estimate of the true paleocolatitude. Values of T depend on two parameters, σ_{SV}, the angular standard deviation of VGP's corresponding to geomagnetic secular variation, and the colatitude, Θ_0, of the sampling site. The heavy curve shows our model of geomagnetic secular variation.

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TABLE 4. Paleocolatitude Corrections T and Standard Deviations S_0 as a Function of Mean Colatitude

Θ, deg	T, deg	S_0, deg
18	Θ	9.6
19	12.7	10.0
20	10.5	10.4
21	9.0	10.8
22	8.0	11.1
23	7.0	11.4
24	6.3	11.6
25	5.8	11.8
26	5.3	12.0
27	4.9	12.2
28	4.4	12.3
29	4.1	12.4
30	3.9	12.5
31	3.6	12.6
32	3.3	12.6
33	3.1	12.7
34	2.9	12.7
35	2.8	12.7
36	2.6	12.7
37	2.4	12.7
38	2.2	12.7
39	2.1	12.6
40	2.0	12.6
41	1.9	12.5
42	1.8	12.5
43	1.7	12.4
44	1.6	12.4
45	1.5	12.3
46	1.4	12.2
47	1.3	12.1
48	1.3	12.0
49	1.2	12.0
50	1.1	11.9
51	1.1	11.8
52	1.0	11.7
53	1.0	11.6
54	0.9	11.5
55	0.9	11.4
56	0.8	11.3
57	0.8	11.2
58	0.7	11.1
59	0.7	11.0
60	0.7	10.9
61	0.6	10.8
62	0.6	10.7
63	0.5	10.6
64	0.5	10.5
65	0.5	10.4
66	0.4	10.3
67	0.4	10.2
68	0.4	10.1
69	0.3	10.0
70	0.3	10.0
71	0.3	9.9
72	0.3	9.8
73	0.3	9.7
74	0.2	9.6
75	0.2	9.5
76	0.2	9.5
77	0.2	9.4
78	0.2	9.4
79	0.1	9.3
80	0.1	9.3
81	0.1	9.2
82	0.1	9.2
83	0.1	9.1
84	0.1	9.1
85	0.1	9.1
86	0.1	9.0
87	0.0	9.0
88	0.0	9.0
89	0.0	9.0
90	0.0	9.0
120	−0.7	10.9
150	−3.9	12.5

lated from the observed data but rather is based on a model of secular variation, only one equation is needed to solve for the single unknown Θ_0.

Confidence Limits

Two options are open in determining confidence limits for paleocolatitudes. The first is to begin with the angular standard deviation of the observed colatitudes. The second is to infer the angular standard deviation of the populations of VGP's that correspond to the observed paleocolatitudes and then determine confidence limits for the parent population of poles. The two types of analysis are intrinsically different in the dimensionality of the appropriate probability distribution functions and the significance of the confidence limits obtained from the analysis. Two coordinates (e.g., colatitude and azimuth) are required to specify VGP's and the appropriate PDF is bivariate, whereas only one coordinate, colatitude, is required for the direct approach and the appropriate PDF is univariate. Because the angular standard deviations and standard errors for the bivariate and univariate cases are different, it is important to select the appropriate analysis for the particular hypothesis being tested. As noted earlier, univariate analysis is appropriate for testing the hypothesis that the mean pole that produced an observed set of paleocolatitudes lies at a particular distance from a site, whereas bivariate analysis is appropriate to test the hypothesis that the mean pole is at a particular location. Bivariate confidence limits are substantially larger than univariate limits, leading to the apparent paradox that a pole with a location that is consistent with a given set of paleocolatitude data may have an angular distance from the sampling site that is inconsistent with the same set of data. The paradox is

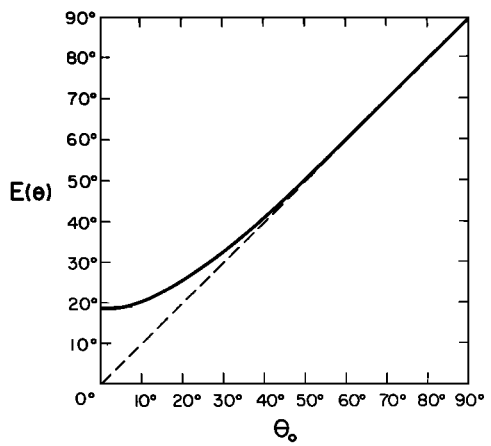


Fig. 7. The expectation, $E(\Theta)$, of the PDF for paleocolatitudes as a function of true paleocolatitude, Θ_0. If $E(\Theta)$ is equated with the mean paleocolatitude, $\bar{\Theta}$, the vertical distance between the solid and dashed curve is the paleocolatitude correction, T, for our model of geomagnetic secular variation.

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more apparent than real because the two hypotheses under consideration are not equivalent.

The main approach in this paper is a direct univariate analysis of colatitudes. If bivariate confidence limits are needed for some application, these can be found using Table 1, which maps values of univariate standard deviation, σ_u, into corresponding bivariate standard deviations, σ_b.

Analysis of Errors

We now discuss the errors that arise in estimating a paleocolatitude Θ^* from paleomagnetic measurements from a single drill site. Errors of four types will be considered: (1) intrinsic paleomagnetic error present after averaging N independent measurements of the varying geomagnetic field, (2) random experimental errors, (3) systematic errors caused by nonvertical drilling or undetected tectonic tilt, and (4) errors in the paleocolatitude correction T. Errors of types 1 and 2, both random, are estimated from N and the variance of the measured colatitudes. Nonvertical drilling and tectonic tilt produce systematic errors of type 3 if, as is usually the case, the orientation of the drill hole or the tilt is unknown but constant for all samples from the hole. Error 4 will be seen to depend on the sum of the other errors.

Random errors of types 1 and 2 are both reflected in the angular standard deviation, S_{0B}, of the N observed colatitudes Θ_i,

$$S_{0B}^2 = \frac{1}{N-1}\sum_i (\Theta_i - \bar{\Theta})^2 \tag{37}$$

Two sources of angular variance contribute to S_{0B}^2. The first is the between-flow variance S_{BF}^2, almost all of which corresponds to variations in the ancient geomagnetic field. The second contribution is from the standard errors of the individual flow means. If S_i is the standard error in colatitude for the ith of N flows as determined from intraflow data, then the corresponding mean variance for the set of N flows is

$$\overline{S^2} = \frac{1}{N}\sum_i S_i^2 \tag{38}$$

This variance adds to the between-flow variance S_{BF}^2 to give the total observed angular variance

$$S_{0B}^2 = S_{BF}^2 + \overline{S^2} \tag{39}$$

This general additive relationship of variances is valid for different types of paleomagnetic data including two-dimensional paleomagnetic directions, two-dimensional VGP's, one-dimensional inclination-only data, and one-dimensional paleocolatitude-only data but is invalid for mixed variances of different type.

The following conversions are useful for converting data to a common form. For two-dimensional data, note the following:

1. If α_i is the 95% confidence cone for the ith direction or VGP of a set of N data, then the corresponding angular standard deviation $\sigma_i = 0.58\alpha_i$.

2. If K is the precision parameter for a set of vectors, then the corresponding angular variance is given by $S^2(\text{deg}^2) = 6566/K$.

3. If the variance S_D^2 is known for a set of paleomagnetic directions, then the variance S_P^2 for the corresponding set of poles obtained from the directions is given by $S_P^2 = R_2 S_D^2$, where

$$R_2 = \frac{2(1 + 3\cos^2\Theta_0)^2}{(5 + 3\cos^2\Theta)} \tag{40}$$

and Θ_0 is the paleocolatitude of the sampling site [*Cox*, 1970]. For example, if each of N magnetic directions has a 95 confidence cone of $\alpha_{95} = 5°$, the corresponding standard error for each is $0.58 \times 5° = 2.90°$. For a site at $\Theta_0 = 0°$, (40) yields a value of $R_2 = 4$. Therefore the mean standard error of 2.90° maps into a variance $\overline{S^2} = 4 \times 2.90^2 = 33.6$ deg^2 for the corresponding VGP's. If for the latter $S_{0B} = 20°$, then $S_{BF}^2 = S_{0B}^2 - \overline{S^2} = 400 - 33.6 = 366.4$ deg^2 and $S_{BF} = 19.1°$.

For one-dimensional data, note the following:

1. Within-flow dispersion is commonly assessed by finding the within-flow standard deviation of inclinations, I_{SD}, of n specimens from one stratum. The standard error of $\bar{I}$ for the ith flow is $s_i = I_{SD}/(n)^{1/2}$ and the mean variance of the mean inclination for all flows is $\overline{s^2} = (1/N)\sum s_i^2$.

2. If a variance $\overline{s^2}$ is known for mean inclinations whereas an analysis is being made of the corresponding paleocolatitudes, $\overline{s^2}$ can be converted to the corresponding colatitude variance $\overline{S^2}$ by $\overline{S^2} = R_1\overline{s^2}$, where [*Cox*, 1970]

$$R_1 = \frac{(1 + 3\cos^2\Theta_0)^2}{4} \tag{41}$$

and Θ_0 is the true paleocolatitude of the site. For example, if inclination-only measurements from six flows have standard errors of 4°, 4°, 5°, 5°, 6°, and 6°, then $(\overline{s^2})^{1/2} = 5.07°$ is the mean standard error of the inclination measurements. To map this into an equivalent variance $\overline{S^2}$ in colatitude for a site at $\Theta_0 = 30°$, by (41), $R_1 = 2.64$ and therefore $\overline{S^2} = 2.64 \times 5.07^2 = 67.9$ deg^2. If the observed standard deviation S_{0B} in colatitudes is 13°, then $S_{BF}^2 = 169 - 67.9 = 101.1$ deg^2 and $S_{BF} = 10.1°$.

A key issue in analyzing paleomagnetic data is assessing whether the between-flow dispersion in paleocolatitudes measured by S_{BF} represents a complete sampling of geomagnetic secular variation. For a site at distance Θ_0 from a mean pole about which VGP's are dispersed with an angular standard deviation σ_{SV}, the expected standard deviation S_0 in colatitudes can be read from Table 4, where values of S_0 are listed at 1° increments of $E(\Theta) \approx \bar{\Theta}$ for the model of secular variation used in the present analysis. Quite commonly on comparing observational data with this table, the observed between-flow standard deviation, S_{BF}, is not significantly different from the standard deviation S_0 predicted by our global SV model. The value of S_0 of the global model is then to be preferred over the observed value of S_{BF} in calculating confidence limits because the former is based on much more data. Using the value S_0 for the global model (Table 4), the best estimate of the total variance S_R^2 of the mean colatitude attributable to random errors is

$$S_R^2 = \frac{S_0^2 + \overline{S^2}}{N} \tag{42}$$

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where N is the number of independent observations.

If $S_{BF} << S_0$, the most likely explanation is that the full range of SV has not been sampled and therefore (42) should be used. Values of $S_{BF} >> S_0$ may have any of three causes: (1) geologic noise, e.g., random tectonic tilting; (2) SV greater than that of the model; and (3) inadequate sampling of the geomagnetic field. For 1 and 2, the best estimate of S_R^2 is provided by S_{0B}^2/N, where S_{0B} is the observed angular standard deviation of colatitudes. For case 3, (42) is appropriate even though this will yield a smaller confidence interval than would the observed angular standard deviation.

Systematic errors caused by deviation of the drill hole from the vertical or tectonic tilt of an entire drilled section do not decrease with an increase in the number N of independent data. If the total variance of systematic errors is S_S^2 and the total variance of random errors is S_R^2, then the total standard error, S_M, in the mean colatitude is given by

$$S_M = (S_R^2 + S_S^2)^{1/2} \quad (43)$$

As an example, at DSDP site 433, the maximum deviation from the vertical of the borehole was 2.5° (W. J. Morgan, personal communication, 1981). Interpreting this conservatively by setting $S_S = 2.0°$, if we assume hypothetical values of $S_0 = 10.6°$, $N = 25$, and $(\overline{S^2})^{1/2} = 3°$, then $S_R^2 = (10.6^2 + 3^2)/25 = 4.85$ and $S_R = 2.20°$. Combining this random error with the systematic error of 2.0°, the total error of the mean is $S_M = (2.20^2 + 2.0^2)^{1/2} = 3.0°$. This example shows that systematic errors, even when small, can contribute significantly to the size of confidence limits. However they are difficult to evaluate because they are not reflected in the dispersion of data.

The standard error of our estimate $\Theta^* = \overline{\Theta} - T(\overline{\Theta})$ of the true paleocolatitude Θ_0 is not simply equal to S_M because errors in $\overline{\Theta}$ introduce errors in the colatitude correction $T(\overline{\Theta})$ which is a function of $\overline{\Theta}$. For small values of $\overline{\Theta}$ the effect is to substantially widen the confidence limit of Θ^* over that of $\overline{\Theta}$ because of the strongly nonlinear relationship between $T(\overline{\Theta})$ and $\overline{\Theta}$. The standard error of Θ^* is thus not simply the interval from $S_{(-)} = \overline{\Theta} - S_M - T(\overline{\Theta})$ to $S_{(+)} = \overline{\Theta} + S_M - T(\overline{\Theta})$, but rather the interval from $S_{(-)} = \overline{\Theta} - S_M - T(\overline{\Theta} - S_M)$ to $S_{(+)} = \overline{\Theta} + S_M - T(\overline{\Theta} + S_M)$. For example, if $\overline{\Theta} = 22°$ and $S_M = 3°$, then $\Theta^* = \overline{\Theta} - T(\overline{\Theta}) = 22° - 8° = 14°$ and limits for Θ^* range from $S_{(-)} = 22° - 3° - T(19°) = 19° - 12.7° = 6.3°$ to $S_{(+)} = 22° + 3° - T(25°) = 19.2°$. The range of uncertainty spanned by the standard error is thus $\Theta^* = 14°^{+5.2°}_{-7.7°}$ rather than $\Theta^* = 14°^{+3°}_{-3°}$, the value obtained if the variation of T with $\overline{\Theta}$ is neglected. At the equator this effect is vanishingly small. Where symmetrical confidence limits are required in an analysis of paleocolatitudes from different sites, we will use Θ^* as an estimate of paleocolatitude and $S = (S_{(+)} - S_{(-)})/2$ as an approximation of the standard error.

Number N of Independent Measurements

The subtle and important question of what number to use for N in (42) is crucial to determining valid confidence limits. Most authors, although agreeing that N should represent the number of "independent" paleomagnetic measurements, define "independent" in different ways. For example, in his benchmark study of the paleomagnetism of lava flows from Suiko Seamount, *Kono* [1980] suggested two criteria. The first is that two successive lava flows are independent if their mean inclinations differ by more than 15°, an angular change which, assuming a rate of secular variation similar to that in recent times, would represent a time gap of at least a few hundred years. Kono's second criterion, which is the one he applied in practice, was the less restrictive condition that successive inclinations be statistically different. Data from adjacent flows that failed this test were merged and averaged. Applying this criterion to paleomagnetic data from Suiko Seamount, successive flows with inclinations that differed by as little as 0.4° were regarded as independent. Similar or less restrictive criteria of independence have been used by many workers.

In the present analysis we adopt the more restrictive view generally used in the statistical analysis of time series that samples of a signal that varies as a function of time are independent only if the time intervals between successive samples are all greater than a coherency time τ_c that represents the longest wavelength present in a hypothetical long, continuous record of the signal. The value of τ_c of the geomagnetic field is thought, with considerable uncertainty, to be of the order of 10^4 years. If a sequence of n lava flows or other strata which provide spot readings of the field formed at times that are all separated by intervals of τ_c or more years, the number of independent observations is $N = n$. On the other hand if some of the n strata formed at more closely spaced times, then $N < n$. Geologic considerations suggest that the latter is likely to be true for sequences of lava flows. Volcanism is usually episodic in nature, intervals of quiescence alternating with intervals during which many volcanic cooling units are erupted in rapid succession. As a result, a large set of n individual cooling units may all be extruded in a time shorter than τ_c. In this case, $N \approx 1$. Uncertainty about the number N of independent observations comprises the largest source of uncertainty in assessing confidence limits for paleocolatitudes from paleomagnetic data for lava flows.

Some of the practical difficulties of estimating N are illustrated in Figure 8. The six panels show different samplings of the same continuously varying time signal representing magnetic inclination. The six small bottom panels display how these data would appear in a typical paleomagnetic study in which the sequence of data points is known but their timing is not. In Figure 8*a* the four data points are all separated by time intervals $t > \tau_c$ and therefore $N = n = 4$. In Figure 8*b* the number of observations has been doubled, yet because adjacent pairs are separated by time intervals $t << \tau_c$, only four of the eight observations are independent ($N = 4$). In Figure 8*c*, although the six observations show no serial correlation, a criterion sometimes used for independence, the variance of the data is much less than that of the geomagnetic field because of the close spacing of the samples. These six data points provide little more information than does the single point 2 in Figure 8*a* and therefore $N = 1$. In Figure 8*d* the five data points are similar to those in Figure 8*c* in being uncorrelated serially, but in Figure 8*d* they are separated by intervals $t > \tau_c$. They are thus independent and

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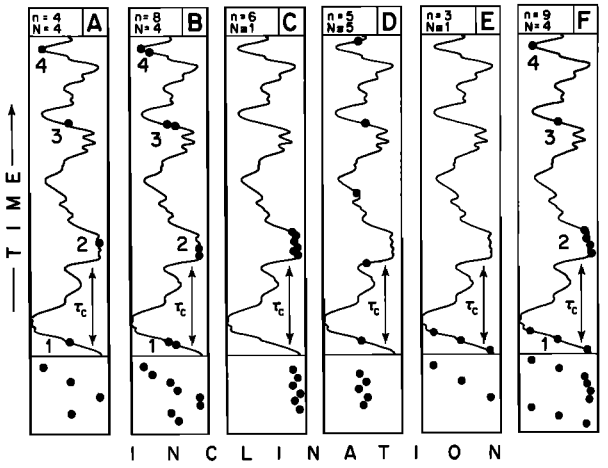


Fig. 8. The six solid curves show hypothetical variations of inclination versus time at six nearby sites on the ocean floor. The solid dots in the upper part of each panel show the times when lava flows recording the geomagnetic field were extruded at each site. The solid dots in the lower panels show hypothetical paleomagnetic inclinations from boreholes at the six sites where, as is usually the case, the sequence between adjacent lavas is known but the elapsed time is not. If the time between successive lavas is less than the coherency time, τ_c, of the geomagnetic variation, then the number, N, of independent observations is less than n, the number of strata.

$N = 5$. The appropriate variance to use in finding confidence intervals for the data in Figures 8*c* and 8*d* is the variance of the geomagnetic field and not that of the data. Note that the data in Figures 8*c* and 8*d* cannot be distinguished observationally in the absence of independent information about their timing. The three data points in Figure 8*e*, which are serially correlated, span a wide range of inclination yet they span only a fraction of the indicated coherency time τ_c, and therefore $N \approx 1$. Note that the three paleomagnetic data points in Figure 8*e* cannot be distinguished from points 2, 3, and 4 in Figure 8*a* in the absence of geological indicators of a hiatus. Figure 8*f* resembles typical inclination data from deep sea drilling sites. With a priori knowledge of the continuous record of the field, it is obvious that the three data of group 1 and the four data of group 2 should be averaged before these two groups are averaged with points 3 and 4. However, the paleomagnetic information data alone (lower panels) do not provide the information required for such a two-tier analysis. This example dramatizes the importance of geologic features like changes in the geochemistry of lavas or the presence of soil horizons that indicate the passage of time. In the absence of such features, the number N of independent observations cannot be evaluated with certainty.

Applications To Data

Paleomagnetic Data from the Bering Sea

Before applying this analysis to inclination-only data from rocks with unknown paleocolatitudes, we will first apply it to a data set for which the inclination, the declination, and the true paleocolatitude are all known. The paleomagnetic data to be analyzed were collected from 84 lava flows on the Pribilof Islands and Nunivak Island in the Bering Sea. Because the flows are all young, their true paleocolatitudes are essentially their present site colatitudes. The lava flows were not sampled in a single stratigraphic sequence on each island but rather at widely spaced outcrops where at most only two or three flows were superimposed. As a result, the entire set of 84 lava flows cannot be placed in an exact ordered sequence according to age. However, on the basis of geomorphology, potassium-argon dating, and reversal stratigraphy, the flows are all known to have formed during the Brunhes normal and Matuyama reversed polarity epochs and thus to span the past 2.5 m.y. Sample localities, polarities, and age relations are described in the work of *Cox et al.* [1966], *Cox and Dalrymple* [1967], and *Hoare et al.* [1968]. Data on magnetic inclination, declination, intensity, and intraflow statistics are given in Table 5. These data provide a basis for examining the validity of the univariate distribution function, the model of secular variation, and the colatitude correction used in the present analysis.

To obtain the accuracy required for this analysis, from 5 to 10 oriented samples were collected from each lava flow. To identify flows requiring magnetic cleaning, at least one sample from each lava flow was progressively ac demagnetized at steps of 12.5, 25, 50, 100, 200, 400, and 800 G peak field. Only in rare cases did the direction of magnetization change progressively. For most of these young lavas, the amount of viscous magnetization is too small to change the direction of magnetization significantly. Where changes in magnetic direction were noted during demagnetization, as in the flow at site P23 (Table 5), all samples from that flow were magnetically cleaned and the cleaned results used in the analysis.

The data from these islands serve to emphasize the importance, when analyzing data from ocean floor drill holes, of determining whether all of the observations are statistically independent, as noted earlier. The observation that magnetic directions determined from nearby sites often are similar suggests that many of the Bering Sea measurements are not independent. On St. Paul Island in the Pribilofs, for example, the flows at sites P22, P27, P28, P46, P50, P51, P52, P55, P56, and P60 all have nearly the same rather unusual direction of magnetization (Figure 9). The flows at sites P40, P43, P44, P45, P47, and P48 have another unusual direction. *Cox* [1971] and *Bogue and Coe* [1981] have shown that the probability of generating such groupings by independent random sampling of the earth's field is vanishingly small. The probable cause of the grouping is that volcanic activity, because of its episodic character, provides a highly irregular sampling of the earth's field. When pulses of volcanic activity occur at times when the geomagnetic field is undergoing unusually large swings caused by secular variation, a pattern like that on St. Paul is produced. On Nunivak Island many of the magnetic directions similarly fall into clusters (Figure 10). These were shown by *Hoare et al.* [1968] to correspond to eruptive episodes. Paleomagnetic data from DSDP cores commonly show a similar pattern of clustering. The result of oversampling the field during times of unusually large SV is to bias the final determination of paleocolatitude.

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Table 5. Paleomagnetic Results from Islands of the Bering Sea

Site No.	Coordinates λ	Coordinates ϕ	Exp	N	$\bar{J}$	I	D	VGP λ	VGP ϕ	α_{95}	Equiv. to	VGC θ
					Pribilof Islands, Brunhes Normal Epoch							
P22	57.21	189.73	NRM	7	2.61	57.5	045.9	54.6	292.2	3.1	P46	(51.9)
P23	57.22	189.70	NRM	8	1.86	19.8	351.0			5.9	*	
"	"	"	200	"	0.90	03.6	346.2	33.5	026.3	2.9	*	(88.2)
P27	57.21	189.66	NRM	7	2.59	55.3	042.6	54.6	298.3	4.0	P46	(54.2)
P28	57.20	189.65	NRM	7	3.77	54.4	044.2	53.1	297.4	3.7	P46	(55.1)
P29	57.20	189.63	NRM	7	10.0	59.4	002.5	72.9	003.1	2.7		49.8
P30	57.21	189.62	NRM	6	14.8	52.6	005.0			5.5		
"	"	"	100	"	10.5	53.6	001.2	67.0	007.2	4.2		55.9
P40	57.16	189.58	NRM	7	5.64	53.8	008.3	66.5	352.2	2.0		55.7
P41	57.16	189.58	NRM	7	5.12	58.2	013.1	69.7	338.9	2.9		51.1
P42	57.16	189.58	NRM	7	9.74	67.3	350.8	81.0	050.7	3.8		39.9
P43	57.16	189.62	NRM	6	4.12	56.9	008.6			5.4	P40	
"	"	"	100	"	3.20	55.8	009.0	68.3	349.6	4.3	"	(53.7)
P44	57.16	189.62	NRM	6	9.22	53.2	011.3			5.7	P40	
"	"	"	100	"	7.15	50.6	011.8	62.9	347.1	4.7	"	(58.7)
P45	57.16	189.62	NRM	6	7.76	51.3	011.2	63.6	347.8	4.0	P40	(58.0)
P46	57.16	189.62	NRM	7	2.14	56.0	035.4	58.9	305.6	2.2		53.5
P47	57.16	189.64	NRM	6	5.56	52.4	011.8	64.6	346.2	1.6	P40	(57.0)
P48	57.16	189.64	NRM	6	9.87	51.3	008.5	64.1	352.9	4.1	P40	(58.0)
P49	57.15	189.65	NRM	6	3.03	69.8	336.5	76.3	094.7	2.2		36.3
P50	57.15	189.65	NRM	6	4.97	56.1	040.0	56.6	300.1	1.6	P46	(53.3)
P51	57.15	189.65	NRM	6	6.19	53.0	042.0	53.2	301.2	3.5	P46	(56.4)
P52	57.13	189.71	NRM	6	2.89	52.4	038.7	54.4	305.5	1.8	P46	(57.0)
P55	57.12	189.72	NRM	7	3.57	54.1	045.0	52.4	296.9	2.0	P46	(55.4)
P56	57.12	189.72	NRM	5	2.69	54.6	039.7	55.6	302.3	1.9	P46	(54.9)
P59	57.12	189.74	NRM	7	8.01	45.8	047.3	45.3	301.3	3.7		62.8
P60	57.12	189.74	NRM	7	2.53	54.5	038.1	56.3	304.2	2.9	P46	(55.0)
P64	57.23	189.90	NRM	7	8.21	76.1	327.0			5.4		
"	"	"	100	"	6.63	76.8	324.2	71.4	138.7	5.3	P65	(25.1)
P65	57.25	189.89	NRM	6	13.7	78.2	322.2	70.2	145.8	4.7		22.7
"	"	"	100	"	11.8	78.4	319.9			5.2		
					Pribilof Islands, Matuyama Reversed Epoch							
P2	56.60	190.20	NRM	7	1.52	59.0	005.6	72.8	355.6	2.0		50.2
P3	56.60	190.20	50	5	3.14	50.5	010.1	63.7	350.5	4.4		58.8
P5	56.60	190.20	NRM	8	1.63	-82.6	149.0	-67.9	350.1	2.4		14.6
P12	56.60	190.20	NRM	8	5.59	-75.7	196.2	-79.7	055.2	3.8		27.0
P13	56.60	190.20	NRM	6	4.36	-66.4	173.0	-81.2	221.6	6.1	**	(41.1)
					Nunivak Island, Brunhes Normal Epoch							
A60	60.21	194.33	NRM	8	2.92	85.2	012.0	69.6	200.00	1.8		9.5
A67	60.06	194.30	NRM	7	1.14	65.1	345.8	74.6	053.4	2.8		42.9

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TABLE 5. (continued)

Site No.	Coordinates λ	Coordinates ϕ	Exp	N	$\bar{J}$	I	D	VGP λ	VGP ϕ	α_{95}	Equiv. to	VGC θ
					Nunivak Island, Brunhes Normal Epoch (continued)							
A69	60.06	194.32	NRM	8	0.80	72.9	008.0	85.6	303.9	3.8		31.6
A70	60.02	194.36	NRM	8	2.63	84.1	151.0	49.4	203.1	3.8		11.7
A71	59.98	194.36	NRM	8	4.08	68.0	322.5	67.4	098.1	2.6		38.9
A72	59.98	194.46	NRM	8	1.92	69.8	345.0	79.7	073.6	2.0	A74	(36.3)
A73	59.96	194.43	NRM	8	4.41	69.3	335.2	74.8	088.1	2.6	A74	(37.1)
A74	59.92	194.40	NRM	8	2.64	69.8	337.8	76.4	086.6	1.1		36.3
A75	59.90	194.38	NRM	8	3.18	66.9	331.3	70.7	084.8	2.5	A74	(40.5)
A76	59.89	194.30	NRM	8	3.38	66.2	334.0	71.2	078.8	1.9	A74	(41.4)
A77	59.90	194.17	NRM	8	11.4	76.1	338.9	79.4	134.2	2.6	A78	(26.3)
A78	59.87	194.17	NRM	7	17.4	75.5	346.1	82.8	132.6	2.4		27.3
A79	59.80	193.89	NRM	8	14.2	77.9	005.7	82.6	211.7	2.0	A81	(23.1)
A80	59.78	193.91	NRM	7	4.57	79.1	003.8	80.7	202.5	3.0	A81	(21.1)
A81	59.78	193.91	NRM	7	7.87	78.2	015.2	79.9	229.2	2.0		22.7
A82	59.76	193.79	NRM	8	6.85	80.4	014.1	77.0	214.1	1.6	A81	(18.7)
A83	59.78	193.79	NRM	6	21.0	80.1	002.9	79.0	198.8	2.0	A81	(19.2)
A86	60.02	193.94	NRM	7	1.03	75.7	026.8	77.0	259.1	9.5	**	(27.0)
A88	60.03	193.91	NRM	7	4.81	83.6	158.0	48.0	201.0	1.9		12.6
A89	60.03	193.92	NRM	8	4.00	84.4	161.9	49.3	199.2	3.4	A88	(11.1)
A90	60.03	193.92	NRM	8	9.09	77.1	174.2	35.5	196.9	2.4		24.6
A91	60.03	193.91	NRM	9	1.81	63.8	018.8	71.7	32.79	2.8		44.5
A92	60.02	193.91	NRM	8	7.18	72.2	021.7	78.4	288.3	4.4		32.7
A96	60.02	193.80	NRM	8	5.59	72.7	064.9	57.9	258.1	2.4		31.9
A107	60.00	193.78	NRM	8	6.41	75.6	328.6	74.8	128.0	2.1		27.2
A115	59.96	193.81	NRM	8	2.38	68.5	006.7	81.0	346.2	1.6		38.2
A116	59.96	193.81	NRM	8	6.07	62.3	010.1	72.5	348.8	3.6		46.4
					Nunivak Island, Matuyama Reversed Epoch							
A28	60.39	193.52	NRM	7	4.68	-77.1	216.7	-72.8	070.9	2.3		24.6
A29	60.38	193.56	NRM	7	0.51	-51.5	201.2	-58.6	157.7	5.5	A30	(57.8)
A30	60.37	193.59	NRM	7	3.88	-50.3	203.8	-56.7	154.6	2.7		58.9
A31	60.36	193.62	NRM	7	1.02	-56.4	200.4	-63.3	155.3	2.7	A30	(53.0)
A32	60.36	193.64	NRM	8	10.4	76.3	58.9	63.3	250.1	3.8		26.0
A33	60.37	193.66	NRM	8	1.45	74.1	013.8	83.2	277.8	2.6	A36	(29.7)
A34	60.37	193.66	NRM	8	17.2	75.0	013.2	83.5	265.8	2.9	A36	(26.2)
A36	60.37	193.68	NRM	7	3.28	74.8	349.0	84.5	120.5	2.4		28.5
A37	60.38	193.69	NRM	8	29.9	75.7	005.8	86.2	238.3	4.1	A36	(27.0)
A38	60.38	193.72	NRM	7	41.8	76.7	350.7	84.0	152.9	3.5	A36	(25.3)
A40	60.39	193.74	NRM	7	11.8	73.4	004.5	87.4	309.1	2.4	A36	(30.8)

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TABLE 5. (continued)

Site No.	Coordinates λ	Coordinates φ	Exp	N	$\bar{J}$	I	D	VGP λ	VGP φ	α_{95}	Equiv. to	VGC θ
					Nunivak Island, Matuyama Reversed Epoch (continued)							
A41	60.39	193.76	NRM	7	9.42	75.1	017.3	81.6	265.2	3.6		28.0
A42	60.39	193.79	NRM	10	3.88	73.7	022.4	78.8	277.8	1.9		30.3
A43	60.39	193.81	NRM	7	9.85	72.4	007.2	83.4	318.3	1.8		32.4
A46	60.38	193.89	NRM	7	3.10	-77.4	122.8	-64.6	320.8	4.4		24.1
A48	60.35	193.92	NRM	8	2.23	-73.7	194.4	-82.8	102.6	1.5		30.3
A50	60.32	193.99	NRM	8	6.94	82.3	342.8	74.2	177.4	2.0		15.1
A51	60.32	194.05	NRM	6	10.8	78.0	078.2	57.1	238.7	3.8		23.0
A52	60.34	194.10	NRM	8	5.12	-83.9	201.3	-71.2	027.8	3.4	A54	(12.1)
A54	60.33	194.22	NRM	8	2.37	-84.2	190.0	-71.5	020.4	2.0		11.5
A55	60.30	194.30	NRM	8	3.13	80.8	343.6	76.6	172.2	2.0		17.9
A57	60.27	194.31	NRM	8	1.55	-61.2	237.3	-51.5	105.6	3.6		47.7
A62	60.16	194.32	NRM	8	7.11	-82.8	223.1	-68.4	041.2	2.1		14.2
A64	60.09	194.33	NRM	5	1.30	-86.7	031.3	-54.4	008.5	4.7		6.6
A65	60.06	194.29	NRM	8	1.32	-78.0	172.3	-82.3	351.3	1.6		23.0
A109	60.00	193.81	NRM	6	2.07	-62.8	172.8	-73.6	212.4	3.0		45.8
A110	60.00	193.81	NRM	8	10.5	-63.5	099.6	-42.2	303.6	5.6	**	(44.9)

Notes

Exp: NRM is natural remanent magnetization. 100 is magnetization after partial demagnetization in peak alternating field of 100 gauss.

N: Number of oriented samples from one cooling unit.

$\bar{J}$: Average intensity of magnetization in units of 10^{-3} emu/cc.

I: Inclination of remanent magnetization.

VGP: Virtual geomagnetic pole, specified by latitude λ and longitude Φ of (-) pole.

α_{95}: 95 percent confidence circle for direction of remanence.

Equivalent to: Flows with nearly the same direction of magnetization as nearby flows are listed here. These were excluded from subsequent analysis to avoid oversampling. *(P23) was also excluded because this flow formed during a time of abnormal field behavior. Flows marked (**) were excluded because the confidence circles were abnormally large.

VGC: Virtual geomagnetic colatitude $\Theta = cot^{\dagger}[(1/2)/\imath n\ I]$. Values in parentheses () were not included in subsequent analysis for reasons indicated in previous column.

To eliminate bias resulting from oversampling flow sequences that formed in rapid succession, the complete data set listed in Table 5 was winnowed by first dropping data from nearby flows with nearly parallel magnetic directions. Next a cutoff of data with 95% confidence limits larger than 5° was made, eliminating three additional data. Finally, the flow at site P23 with a VGP 33.5° from the equator was dropped because this magnetic direction, although undoubtedly a valid paleofield vector, probably represents an excursion or polarity transition of the field rather than normal secular variation [*Doell and Cox*, 1972; *Dagley and Lawley*, 1974; *Harrison*, 1980]. Because the earth's field has quite different characteristics during polarity transitions and excursions, the field directions sampled during such times constitute a statistically distinct population and may appropriately be omitted from analyses of paleofields for the purpose of determining paleolatitudes and paleopoles. The cutoff value of VGP latitudes to use

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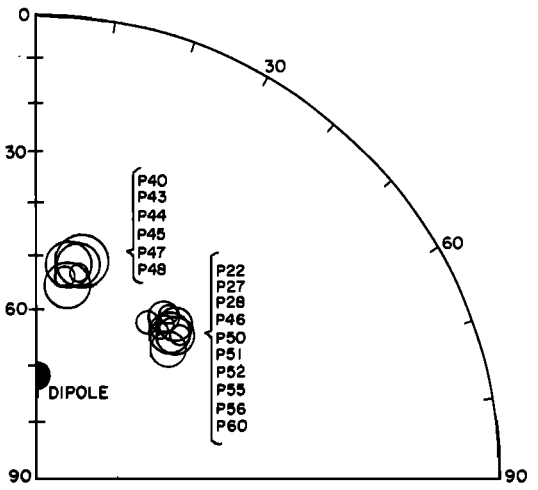


Fig. 9. Paleomagnetic field directions from 16 sites on St. Paul Island, Pribilof Islands, in the Bering Sea. Each circle is a 95% confidence region based on paleomagnetic measurements of six to eight oriented samples from one site. These 16 lava flows appear to have sampled the geomagnetic field at only two independent times. Accordingly only two of the field vectors are used in subsequent analysis of data from the Bering Sea.

as a criterion in recognizing transitions has been estimated to be 40 to 50 degrees [*Doell and Cox*, 1972; *Dagley and Lawley*, 1974; *Harrison*, 1980]. A second criterion is that the intensity of the field commonly decreases by a factor of 3 to 5 during a polarity transition. An unusually low intensity of magnetization at site P23 is consistent with this datum being a transitional field direction.

The final data set of 47 paleofield directions (large symbols, Figure 10) comprise a sample of the earth's field over the Bering Sea during the past 2.5 m.y. that is reasonably unbiased, although even these 47 observations may not be completely independent as suggested by a weak correlation between the spatial distribution of several sites and their magnetic directions.

The angular dispersion of the 47 VGP's for these data (Table 6) has a value of $K = 13.6$, which is significantly smaller than the value of $K_{SV} = 18$ used in our model at colatitude $\Theta_0 = 30°$, as determined at 95% confidence using a one-tailed F ratio test [*McFadden*, 1980]. On the other hand, a value of $K = 19.5$ found by *Bingham and Stone* [1972*b*] for lava flows from the nearby Aleutian Islands at colatitude 37° agrees well with our model for SV. The failure of the Bering Sea data to pass the F ratio test supports the view that even after severe winnowing, the $N = 47$ data may not have been independent. The difference in the dispersions of the Bering Sea and Aleutian Island data sets, which reflects the circumstance that several of the Bering Sea lavas were erupted during times of unusually large SV whereas none of the Aleutian lavas were, serves to emphasize the difficulty of evaluating the paleosecular variation parameter K_{SV} from measurements at a single site.

The fit of the observed paleocolatitudes to the PDF of (5) is shown in Figure 11. The chi-square value of the χ^2 statistic for class intervals 0°–20°, 20°–30°, 30°–40°, and >40° is 1.51. With 4 degrees of freedom because all parameters of the PDF are fixed, the value of $\chi^2 = 1.51$ falls midrange between probability $0.90(\chi^2 > 1.064)$ and $0.80(\chi^2 > 1.649)$, indicating an acceptable fit.

These results provide an opportunity to test the validity of several of the theoretical relationships we have derived linking univariate and bivariate data. The observed univariate standard deviation of colatitudes, S_u, is 14.9°. To estimate the angular standard deviation, S_b, of the corresponding bivariate distribution of VGP's, we note from Table 1 that at the approximate site colatitude, $\Theta_0 = 30°$, a colatitude angular standard deviation of $S_u = \sigma_u = 14.5°$ corresponds to a VGP angular standard deviation $S_b = 22.5°$. Interpolating the table more accurately, the observed value $S_u = 14.9°$ gives $S_b = 23.2°$. The observed angular standard deviation of VGP's from their mean may be estimated from the one-level value of K as $81°/(K)^{1/2} = 81°/(13)^{1/2} = 22.5°$, in good agreement with the estimate of 23.2° made using colatitudes only. Similarly, the observed angular standard deviation of colatitudes, S_u, may be used to estimate the 95% confidence limit for the VGP's from the relationship derived earlier that $\alpha_{95} \approx 1.72(S_b/S_u)S_u/(N)^{1/2} = 1.72 \times 1.55 \times 14.9°/(47)^{1/2} = 5.8°$. Again this is in good accord with the value of $\alpha_{95} = 6.0°$ estimated from a one-tier analysis of VGP's.

As noted earlier, the mean of a set of observed paleocolatitudes is a biased estimator of the true paleocolatitude. For the Bering Sea lavas, the true paleocolatitude as measured by the mean site colatitude is 30.7°. The mean VGC, $\bar{\Theta}$, of the 47 data is 32.9°, which is 2.2° greater than the true colatitude. From Table 4 the paleocolatitude correction T is 3.1°, which slightly overcorrects $\bar{\Theta}$, but by an amount that we will see is well within experimental error. By (36) the best estimate of the colatitude on the basis of the paleomagnetic data is given by $\Theta^* = \bar{\Theta} - T = 29.8°$.

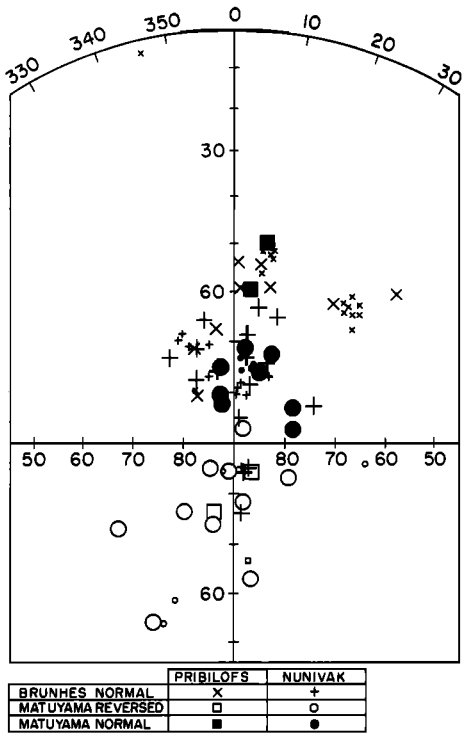


Fig. 10. Paleomagnetic field vectors from lava flows sampled on the Pribilof Islands and Nunivak Island in the Bering Sea. Large Symbols indicate Data from sites assumed to provide an independent sample of the geomagnetic field. Small Symbols indicate Data from sites with ages so close to those of the independent sites that they do not provide an independent measurement of the geomagnetic field.

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TABLE 6. Summary of Paleomagnetic Results from 47 Bering Sea Lava Flows

	One Level	Two Level
Brunhes normal epoch (N = 25)		
Mean VGP latitude	86.3	
Mean VGP longitude	270.0	
R	22.7834	
K	11.0	11.0
α_{95}	9.2	8.5
s.d. of VGP's from axial dipole	25.4	25.2
Matuyama reversed epoch (N = 22)		
Mean VGP latitude	8.18	
Mean VGP longitude	240.8	
R	20.7126	
K	16.0	18.2
α_{95}	7.9	7.1
s.d. of VGP's from axial dipole	21.9	21.7
Brunhes and Matuyama epochs (N = 47)		
Mean VGP latitude	84.3	
Mean VGP longitude	250.4	
R	43.4505	
K	13.0	13.6
α_{95}	6.0	5.6
s.d. of VGP's from axial dipole	23.5	23.4
Mean VGC	32.9	
s.d. of VGC's	14.9	
Mean site colatitude	30.7	

To determine confidence limits for this "paleocolatitude," the starting point is the angular standard deviation, S_{0B}, of the 47 colatitudes, which is 14.9°. To determine what part of this is contributed by intraflow errors, the values of the bivariate 95% confidence interval, α_i, as listed in Table 5, were first converted to standard errors by $s_i = 0.58\alpha_i$, where s_i is the standard error of the ith paleomagnetic direction. The mean variance in inclination for all flows is $\overline{s^2} = 1/n\sum s_i^2 = 2.95$ deg^2. This can be converted to the corresponding colatitude variance, $\overline{S^2}$, by $\overline{S^2} = R_1\overline{s^2} = 7.8$ deg^2 (equation (41)). The between-flow angular standard deviation is given by $S_{BF} = (S_{0B}^2 - \overline{S^2})^{1/2} = (14.9^2 - 7.8)^{1/2} = 14.6°$. At the mean colatitude of the sites, $\bar{\Theta} = 29.8°$, the expected angular standard deviation, S_0, owing to geomagnetic secular variation is 12.5°. The 2.1° difference between S_{BF} and S_0 is of doubtful significance and uncertain origin. If it reflects inadequate sampling of the geomagnetic field, then as discussed earlier the best estimate of the standard error of the mean inclination is $S_R = [(S_0^2 + \overline{S^2})/N]^{1/2} = 1.9°$ (equation (42)). On the other hand, if the difference between S_{BF} and S_0 reflects tilting or other geologic noise, then the best estimate of the standard error is $S_R = [S_{0B}^2/N]^{1/2} = 2.2°$. Assuming that the former is correct and that there are no systematic errors, then the standard error of the mean colatitude, S_M, is given by $S_M = S_R = 1.9°$ (equation (43)). The lower bound, $S_{(-)}$, of the confidence interval for the mean colatitude is given by

$$S_{(-)} = \bar{\Theta} - S_M - T(\bar{\Theta} - S_M)$$
$$= 32.9° - 1.9° - T(31°)$$
$$= 31° - 3.6° = 27.4°$$

For the upper bound,

$$S_{(+)} = \bar{\Theta} + S_M - T(\bar{\Theta} + S_M)$$
$$= 32.9° + 1.9° - T(34.8°)$$
$$= 34.8° - 2.8° = 32.0°$$

The range of uncertainty corresponding to the standard error is thus $\Theta^* = 29.8°^{+2.2°}_{-2.4°}$, which reassuringly encompasses the mean site colatitude of 30.4°. If Θ^* were to be used as input in a subsequent analysis of colatitudes, then as noted earlier the appropriate mean standard error would be 2.3° rather than $S_M = 1.9°$.

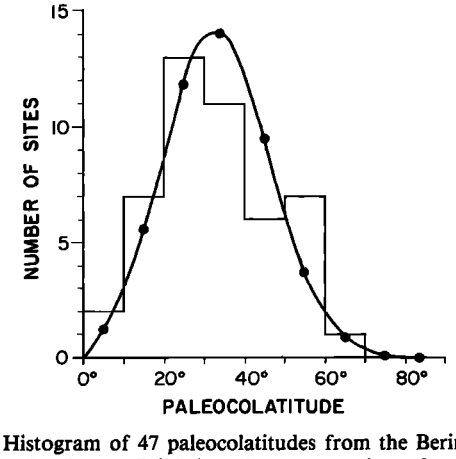


Fig. 11. Histogram of 47 paleocolatitudes from the Bering Sea for class intervals of 10°. Solid dots show expected number of sites in each 10° interval based on (5) and our model of geomagnetic secular variation. The χ^2 statistic (class intervals 0°-20°, 20°-30°, 30°-40°, >40°, $\chi^2 = 1.51$, four degrees of freedom) falls midrange between probability 0.90 ($\chi^2 > 1.064$) and probability 0.80 ($\chi^2 > 1.649$), indicating an acceptable fit.

DSDP Leg 55 Basalts

The classic study of the paleomagnetic inclinations of cores from Ojin, Nintoku, and Suiko seamounts by *Kono* [1980] provides the best available basis for comparing the present method of analysis with that of *Kono* [1980]. *Kono*'s [1980] approach is to equate the expectations of $\cos\Theta$ and $\cos^2\Theta$ to the observed values with the goal of obtaining an unbiased estimate, Θ_K, of the true colatitude

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TABLE 7. Paleomagnetic Data From Suiko Seamount and Midway Island

Site	433	433*	Midway	Midway*
N	65	40	13	8
$\bar{\Theta}$	63.4	63.7	74.7	73.7
Θ^*	62.9	63.2	74.5	73.5
Θ_K	(62.9)	(63.1)	(74.6)	(73.2)
S_{0B}	11.1	11.3	7.6	7.3
S_b	15.7	16.0	10.7	9.9
S_K	(15.3)	(15.9)	(10.3)	(10.0)
α_{95}	3.3	4.3	5.1	6.3
α_K	(3.4)	(4.5)	(5.4)	(6.9)

$\bar{\Theta}$, mean colatitude; Θ^*, Θ corrected for inclination error assuming present SV; Θ_K, estimate of colatitude by method of Kono (1980); S_{0B}, standard deviation of colatitude data; S_b, $\sqrt{2}\,S_{0B}$; S_K, estimate of standard deviation of VGP's [*Kono*, 1980]; α_{95}, $2.43S_{0B}/\sqrt{N}$; α_K, estimate of α_{95} of VGP's from Table 6 of *Kono* [1980] with errors corrected.

* Where adjacent flows have similar directions, data have been grouped.

and an estimate, K_K, of the precision parameter of the bivariate Fisherian distribution of VGP's corresponding to the observed set of colatitudes. Using Monte Carlo experiments, *McFadden and Reid* [1982] found that estimates of Θ_K found by this method may be biased if Θ_0 is small and K is large. From K_K *Kono* [1980] calculates the angular standard deviation, $S_K = 81°/(K_K)^{1/2}$, of the VGP distribution and the radius of the 95% confidence circle,

$$\alpha_K = \cos^{-1}\left[1-\frac{N-1}{(K_K-1)N+1}\left[20^{1/(N-1)}-1\right]\right] \quad (44)$$

Table 7 lists the value of these parameters for Suiko Seamount and Midway Island from *Kono* [1980], where the numbers in parentheses and those listed immediately above them describe parameters obtained by Kono's method and ours, respectively. For example, for the $N = 40$ grouped data from Suiko Seamount (site 433), from the observed univariate standard deviation of the colatitudes, $S_{0B} = 11.3°$, we estimate that the bivariate standard deviation of the corresponding VGP's is $S_b = (2)^{1/2}S_{0B} = 16.0°$ whereas Kono's estimate is $S_K = 15.9°$. Similarly our estimate for the 95% confidence limit is $\alpha_{95} = 2.43S_{0B}/(N)^{1/2} = 4.3°$, whereas Kono's estimate (after a minor correction in his Table 6) is $\alpha_K = 4.5°$. These results are encouraging because both approaches yield similar estimates for the parameters of the hypothetical bivariate VGP population that produced the observed colatitudes.

The main difference between our analysis and that of *Kono* [1980] is in the approach used to assess the number, N, of independent observations. Kono regarded data from adjacent flows as being independent if their inclinations are significantly different from each other, even if successive inclinations lie along a smooth curve and are obviously serially correlated. In our analysis, points are regarded as independent only if their spacing in time is greater than the largest period present in geomagnetic secular variation; therefore we would regard points lying along a smooth curve as not independent. Of the 65 inclinations determined from individual basalt flows on Suiko, our estimate of the number of independent data points is $N = 20$, whereas Kono estimates $N = 40$.

Turning to a detailed analysis of errors for the Suiko data, the observed standard deviation of the colatitudes for 40 grouped flows is $S_{0B} = 11.3°$. The standard error in colatitude, $(\overline{S^2})^{1/2}$, corresponding to the values of α_{95} listed by *Kono* [1980] for inclinations is 1.20°. Therefore by (39) the between-flow standard deviation, S_{BF} is $(11.3° - 1.2°)^{1/2} = 11.2°$. The standard deviation, S_0, predicted by our model for secular variation at mean colatitude $\bar{\Theta} = 63.7°$ is 10.6°, which is not significantly different from $S_{BF} = 11.2°$. By (42) the best estimate of the standard error of the mean colatitude due to random errors is thus $S_R = [(S_0^2 + \overline{S^2})/N]^{1/2} = [(10.6^2 + 1.2^2)/20]^{1/2} = 2.4°$. The nature and magnitude of systematic errors owing, for example, to regional tilt or deviation of the bore hole from the vertical, are not fully known. The maximum logged value of the latter was 2.5° (W. J. Morgan, personal communication, 1981). If we estimate the total systematic errors, S_S, to be equal to 2.0°, then the total standard error of the mean S_M, is given by $S_M = (2.4^2 + 2.0^2)^{1/2} = 3.1°$. If at the beginning we had used the observed standard deviation of colatitudes, $S_{0B} = 11.2°$, rather than $S_0 = 10.6°$, then the final standard error of the mean would have been 3.2° rather than 3.1°, a negligible difference.

From Table 4 the colatitude correction for a mean colatitude $\bar{\Theta} = 63.7°$ is $T(63.7°) = 0.5°$, yielding as the best estimate of the true colatitude $\Theta^* = \bar{\Theta} - T(\bar{\Theta}) = 63.7° - 0.5° = 63.2°$. The standard error of Θ^* is the interval from $S_{(-)} = \bar{\Theta} - S_M - T(\bar{\Theta} - S_M)$ to $S_{(+)} = \bar{\Theta} + S_M - T(\bar{\Theta} + S_M)$ or from $S_{(-)} = 63.7° - 3.1° - T(60.6°) = 60.0°$ to $S_{(+)} = 63.7° + 3.1° - T(66.8°) = 66.4°$. The final best estimate of the colatitude is $\Theta^* = 63.2°^{+3.2°}_{-3.2°}$. At this low latitude the effect of the nonlinearity of the colatitude correction is simply to increase the standard error from 3.1° to 3.2°. The bivariate 95% confidence limit for the VGP's corresponding to this set of colatitude data is given by $\alpha_{95} \approx 1.72 \times (2)^{1/2} \times 3.2° = 7.8°$. Comparing this with *Kono*'s [1980] value (after correcting for a trivial error) of 4.5°, the difference reflects our estimate of $N = 20$ rather than his of $N = 40$ and our incorporation of a systematic standard error of 2°. Even with our larger confidence limits, the Suiko paleocolatitude differs from the present colatitude of 70.5° of the Hawaiian hotspot by more than two standard errors, supporting the conclusion that since the time of Suiko the earth's rotation axis has moved farther from the hotspot [*Kono*, 1980].

BOREHOLE DATA FROM THE PACIFIC PLATE

An important conclusion of the present analysis is that estimates in the literature of the number N of independent data are generally too large, with the result that confidence limits are too small. In this section we will test this conclusion by analyzing all of the inclination-only paleomagnetic data from cores drilled into basaltic crust of Cretaceous age on the Pacific plate. These data will be analyzed in two different ways. In the first we will assume that N is equal to n, the number of basalt flows or intrusions and that the standard error is $S_{0B}/(n)^{1/2}$, where S_{OB} is the observed between-flow scatter of colatitudes. In the second we will assume that N is sometimes less than n and that the standard error is $S_0/(N)^{1/2}$, where S_0 is predicted by our model for secular variation. The results of

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both analyses will then be used as input to a second level of analysis that finds the best fit paleomagnetic pole for paleocolatitudes from different sites. The best fit criterion is minimization of the sums of squares of weighted residuals, each residual being the misfit between a site colatitude and the paleomagnetic pole. If the confidence limits assigned to the input data are correct, the residual differences between the input data and the final model will be comparable in magnitude to the confidence limits of the input data. We will see on the basis of chi-square tests that the confidence limits based on the assumption that $n = N$ are much too small.

Review and Analysis of Colatitude Data from the Pacific Plate

The raw data used in our analysis are the lists of paleomagnetic inclinations published in DSDP volumes. In Table 8 are listed the site coordinates, the number n of flows or intrusive cooling units sampled, our assessment of the number N of independent observations, and the normal (N) or reversed (R) polarity of the unit, the latter based either on mutual consistency with other paleomagnetic data from the Pacific plate or on stratigraphic control adequate to tie the site to a specific normal or reversed polarity interval. The latter is especially useful for basalts that formed during the long normal polarity interval that extended from early Aptian time to the end of Santonian. The ages listed in Table 8 are those assigned in the original DSDP volumes or subsequent publications. The paleocolatitude of each cooling unit used in our analysis is shown in Figure 12.

The results of three different analyses made under the three following different sets of assumptions are listed in Table 9.

Analysis A ($N = n$). The inclination of each cooling unit is given equal weight and converted to a colatitude; the mean colatitude is found as the mean of the n colatitudes; the number N of independent observations is assumed to be equal to n, the number of cooling units; no $T(\bar{\Theta})$ correction is made for the inclination bias; the standard error is $S_{0B}/(N)^{1/2}$, where S_{0B} is the observed between-flow scatter.

Analysis B ($N \leqslant$ n, no grouping). The mean colatitude is the mean of the n observations; a value of $N < n$ is used for the number of independent data, where N is based on a subjective assessment of geologic indicators of hiatuses between cooling units and on the basis of the serial correlation of inclinations; the $T(\bar{\Theta})$ correction is made for the colatitude bias; the standard error is $S_0/(N)^{1/2}$.

Analysis C ($N \leqslant$ n, grouping). Groups of data from successive cooling units with obvious serial correlation are first averaged and then analyzed in the same way as the ungrouped data in analysis B; N is the same as it is in analysis B; $T(\bar{\Theta})$ correction is made; the standard error is $S_0/(N)^{1/2}$.

In finding intragroup or intrasite mean colatitudes, data were given equal weight regardless of the standard error of each datum. The rationale for this is that each datum represents an independent point in time, and angular differences between independent paleomagnetic measurements of the geomagnetic field are generally larger than measurement errors. The procedure used in the analysis consisted of the following set of calculations.

1. For each cooling unit, convert the inclination I_i and standard error s_i to the corresponding mean colatitude Θ_i (equation (1*b*)) and the associated standard error S_i (equation (41)).
2. Where only one inclination is available from a cooling unit, estimate S_i as the rms intraunit standard deviation (not error) of the cooling units from the same site.
3. For analysis C only, group together m successive colatitudes that are serially correlated, find the mean group colatitude Θ_i, and estimate the group standard error $S_i = [(S_1^2 + S_2^2 + \cdots + S_m^2)/m]^{1/2}$.
4. Find the mean variance for the site (equation (38)), $\overline{S^2} = (1/N)\sum S_i^2$.
5. Calculate the mean site colatitude $\bar{\Theta} = \sum\Theta_i/N$.
6. For analyses B and C, use Table 4 to find $T(\bar{\Theta})$ and make the colatitude correction to find the unbiased colatitude estimate, $\Theta^* = \bar{\Theta} - T(\bar{\Theta})$.
7. Calculate the standard deviation of the colatitudes, $S_{0B} = [\sum(\Theta_i - \bar{\Theta})^2/(N-1)]^{1/2}$.
8. For analysis A, set standard error $\sigma = S_{0B}/(n)^{1/2}$.
9. For analyses B and C, calculate the between-flow standard deviation, $S_{BF} = (S_{0B}^2 - \overline{S^2})^{1/2}$.
10. Compare S_{BF} with the value of S_0 predicted at colatitude Θ^* (Table 4). Values of S_{BF} much larger or smaller than S_0 suggest that $N < n$.
11. Calculate random error, $S_R = (S_0^2 + \overline{S^2})^{1/2}$.
12. Estimate systematic error, S_S, due to regional tectonics, deviation of the borehole from the vertical, or departure of the geomagnetic field from the configuration of an axial dipole. In the absence of specific information, we estimate that the standard error in inclination is 2° and convert this to the standard error in colatitude by (41).
13. For analyses B and C, calculate standard error, S_M, of the mean colatitude, $\bar{\Theta}$. $S_M = (S_R^2 + S_S^2)^{1/2}$.
14. Map $\bar{\Theta} \pm S_M$ into an interval about Θ^* extending from $\bar{\Theta}_{(-)} = \bar{\Theta} - S_M - T(\bar{\Theta} - S_M)$ to $\bar{\Theta}_{(+)} = \bar{\Theta} + S_M - T(\bar{\Theta} + S_M)$. This is the 68% confidence interval for the unbiased estimate, Θ^*, of colatitude.
15. Estimate the standard error, σ, of Θ^*: $\sigma = (\Theta_{(+)} - \Theta_{(-)})/2$.

A comparison of analyses A and B in Table 9 shows that, as expected at these low-latitude sites, the colatitude correction is negligible. However, the standard errors can be seen to depend strongly on assumptions about N. The sensitivity of mean colatitudes to two-tier grouping may be seen by comparing analysis B (ungrouped) with analysis C (grouped): shifts of the mean upon grouping are all smaller than the standard errors. Our best estimates of the statistical parameters for the paleomagnetic borehole data from the Pacific plate, based on analysis C, are listed in Table 9. The final results of this analysis are the two columns listing the unbiased estimate of the mean colatitude, Θ^*, and its associated error, σ, both of which are used in the next stage of the analysis.

Cretaceous Paleomagnetic Pole

The Cretaceous paleomagnetic data in Table 9 were divided into two subsets, one of Early Cretaceous and one of Late Cretaceous age. Describing the values of Θ^* and σ listed in Table 9 by the new symbols Θ_j and σ_j, these pa-

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TABLE 8. Site Data

Site	Site Location °N Lat, °E Lon	Age	Polarity	n	N	Comment	Reference
Midway Reef Hole	28.3, -177.3	28 Ma	N	10	5	Three sets of flows grouped on basis of similar directions.	Gromme and Vine, 1972
Midway Sand Island Hole	28.3, -177.3	28 Ma	N/R	3	2	Top 2 flows grouped on basis of similar direction; reversal in lowest flow.	Gromme and Vine, 1972
57.0	8.7, 143.5	Late Oligocene	N(?)	1	1		Marshall, 1978; Fischer et al., 1971
61.0	12.1, 147.1	Sant.	N	1	1		Marshall, 1978; Winterer et al., 1971
63.0	0.8, 147.9	Early Oligocene	N(?)	2	1	Flow 1 rejected because of large experimental error.	Marshall, 1978; Winterer et al., 1971
163	11.2, -150.3	Camp.	R(?)	6	2	n < N because of serial correlation, small scatter, etc.	Marshall, 1978; Van Andel, et al., 1973
164	13.2, -161.5	Barr.-Albian	N	12	3	n < N because of serial correlation, small scatter, and apparent absence of hiatuses.	Marshall, 1978; Winterer et al., 1973
165a	8.2, -164.9	Camp.	R(?)	2	1	Inclinations from top flow are divergent and were rejected. Bottom flow may be multiple.	Cockerham, 1979, pers. comm.; Winterer et al., 1973
166	3.8, -175.1	Haut.	R	4	2	Lower 3 units grouped because directions are similar.	Cockerham, 1979, pers. comm.; Winterer et al., 1973
167	7.1, -176.8	Tith.-Berri.	R	4	3	Top 2 units grouped because directions are similar.	Cockerham, 1979, pers. comm.; Winterer et al., 1973
192a	53.0, 164.7	Maas.	N	6	2	n < N because of small scatter and high serial correlation.	Marshall, 1978; Creager et al., 1973
289	-0.5, 158.5	Aptian	N	1	1		Hammond et al., 1975; Andrews et al., 1975
303a	40.8, 154.5	M-4	N	3	1	All units grouped because directions are similar.	Larson and Lowrie, 1975
304	39.3, 155.1	M-7	R	1	1		Larson and Lowrie, 1975
307	28.6, 161.0	M-21n	N(?)	5	3	Flows grouped on basis of similar directions.	Larson and Lowrie, 1975
313	20.2, -171.0	Camp.	N(?)	2	1	Flows grouped on basis of similar directions.	Larson and Lowrie, 1975; Larson et al., 1975
315a	4.2, -158.5	94 Ma	N	6	4	Only one major break in inclination.	Cockerham and Hall, 1976; Cockerham, 1979, pers. comm.; Schlanger et al., 1976
317a	-11.0, -162.3	Barr.-Apt.	R	10	7	Similar directions were grouped in the absence of geologic evidence for a hiatus.	Cockerham, 1979, pers. comm.; Schlanger et al., 1976
430a	38.0, 170.6	55 Ma	R/N	5	3	Flows grouped on basis of similar directions. Inclination of lowest flow opposite from others.	Kono, 1980; Jackson et al., 1980
432a	41.3, 170.4	56 Ma	R	3	2	Flows grouped on basis of similar directions.	Kono, 1980; Jackson et al., 1980
433c	44.8, 170.0	65 Ma	R	65	20		Kono, 1980; Jackson et al., 1980
462	7.2, 165.0	< Ceno. Camp?	N	3	2	n refers to groups of flows with similar petrology.	Steiner, 1981; Larson et al., 1981
462a(x)	7.2, 165.0	Barr.	N	6	4	n refers to sets of flows grouped on basis of magnetic directions and style of extrusion.	Steiner, 1981; Larson et al., 1981
462a(i)	7.2, 165.0	< Ceno Camp?	N	19	9	n refers to individual sills and groups of sills; breccia results excluded.	Steiner, 1981; Larson et al., 1981
465a	33.8, 178.9	Late Albian	N(?)	15(a)	3	Top six samples excluded because magnetic intensities were anomalously low.	Sayre, 1981; Larson et al., 1981

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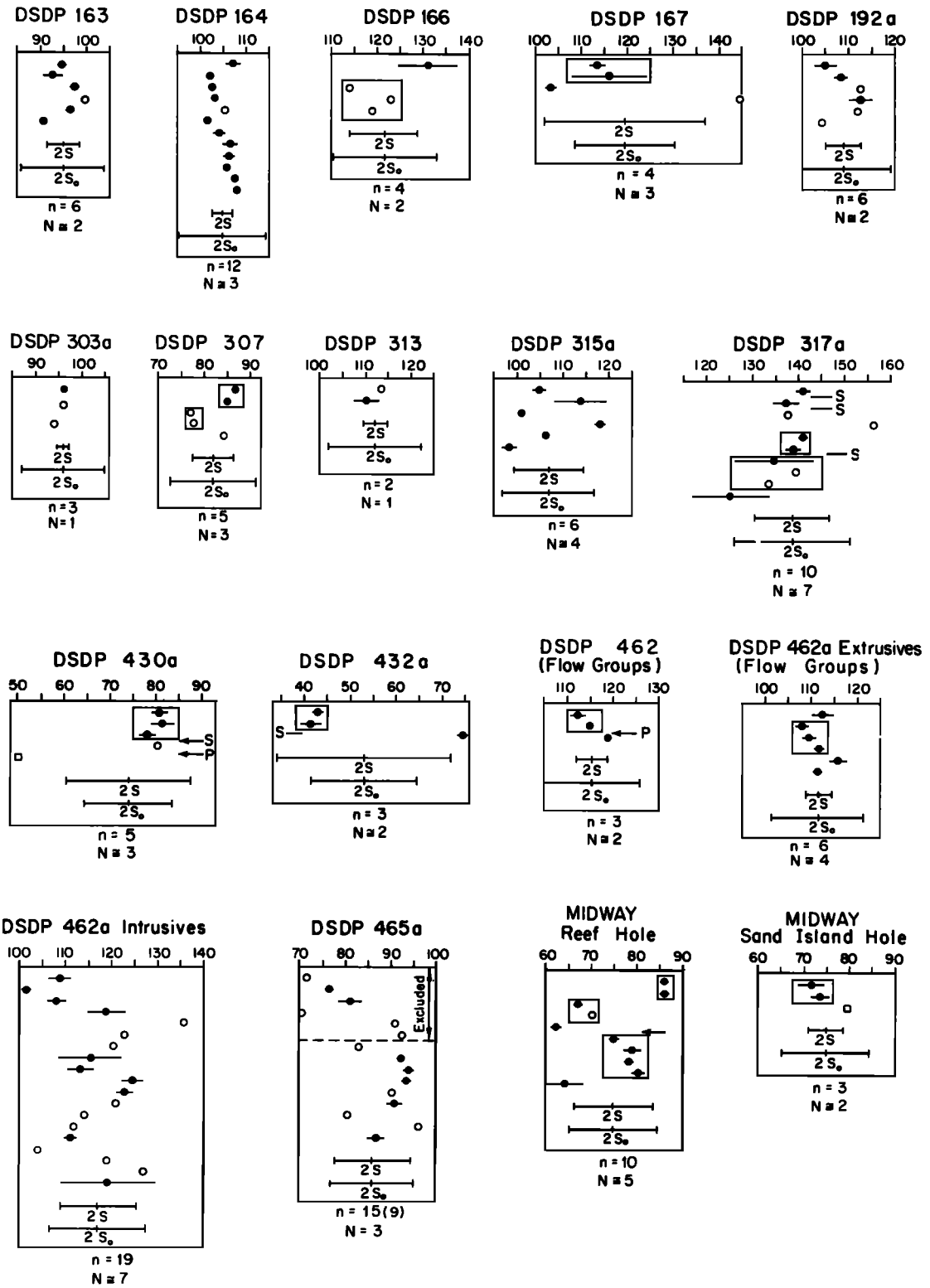


Fig. 12. Cretaceous paleocolatitudes from boreholes in the Pacific basin. Solid circle and associated horizontal bar indicate mean paleocolatitude and standard error from stratum with multiple samples. (Some standard error bars lie within solid circles.) Open circle indicates paleocolatitude from stratum with only one sample. Open square indicates paleocolatitude from a stratum with polarity reversed relative to overlying strata. Other labels are as follows: $2S$, length of line is twice the standard deviation, S, of the observed site paleocolatitudes; center of line is at $\bar{\Theta}$ and ends are at $\bar{\Theta} \pm S$; $2S_0$, length of line is twice the standard deviation, S_0, predicted by our model of geomagnetic secular variation; center of line is at $\bar{\Theta}$ and ends are at $\bar{\Theta} \pm S_0$; P, Hiatus indicated by petrologic variation; S, Hiatus indicated by presence of sediments. Paleocolatitudes within the box were averaged for Analysis C. Parameter n is the number of measured paleocolatitudes; N is the estimated number of statistically independent measurements of the geomagnetic field.

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TABLE 9. Best Estimate of Colatitude, Θ^*, and Angular Standard Error of the Mean, σ

Site	Analysis A		Analysis B		Analysis C	
	Θ^*	σ	Θ^*	σ	Θ^*	σ
Reef Hole	74.7	2.7	74.5	4.7	71.3*	4.9
Sand Island	74.8	2.1	74.6	7.2	75.7*	7.2
57	88.1		88.1	9.3	88.1	9.3
61	115.8		116.3	10.9	116.3	10.9
63	93.9		94.0	9.3	94.0	9.3
163	95.2	1.4	95.3	6.6	95.3	6.6
164	105.1	0.6	105.3	5.8	105.3	5.8
165a	96.2		96.3	9.3	96.3	9.3
166	121.8	3.6	122.5	9.8	125.7*	9.3
167	119.3	8.7	119.9	7.5	121.5*	7.7
192a	70.8	1.6	70.5	7.4	70.5	7.4
289	122.5		123.3	11.5	123.3	11.5
303a	95.7	0.6	95.8	9.2	95.8	9.2
304	101.3		101.4	9.6	101.4	9.6
307	82.1	2.0	82.0	5.6	82.0	5.6
313	111.9	1.6	112.3	11.0	112.3	11.0
315a	107.1	3.1	107.4	5.3	107.4	5.3
317a	138.4	2.5	140.3	6.6	140.9*	6.6
430a	74.1	6.0	73.9	5.9	69.9*	6.1
432a	52.9	11.0	51.9	8.8	57.7*	8.3
433c	63.7	1.5	63.2	3.2	63.2	3.2
462	115.4	2.0	115.9	7.8	116.9*	7.9
462a ext.	111.6	1.1	111.9	5.5	112.8*	5.6
462a int.	116.8	1.9	117.3	4.6	117.3	4.6
465a	89.8	1.7	89.8	5.5	89.8	5.5

* Grouped data.

leocolatitudes and their standard errors are now used as input in a calculation to determine Early and Late Cretaceous poles. In this analysis the pole is found that minimizes the quantity

$$\chi^2 = \sum_{j=1}^{M} \frac{(\Theta_j - \Theta_j')^2}{\sigma_j}^2 \tag{45}$$

where Θ_j is the observed paleocolatitude for the jth site, Θ_j' is the paleocolatitude at the jth site predicted by the model pole, σ_j is the standard error of Θ_j, and M is the number of sites. Minimization is done by a nonlinear least squares technique in which the standard errors, σ_j, are linearly propagated through the calculation to obtain an ellipse of confidence for the final pole [*Gordon and Cox*, 1980]. If the input standard errors, σ_j, are too small by a certain factor, then the final error ellipse for the pole will be too small by the same factor. The final results of this analysis are the latitude, λ, and the longitude, ϕ, of the best fit pole and the variance-covariance matrix of λ and ϕ [*Gordon and Cox*, 1980]

$$W = \begin{bmatrix} \sigma_{\lambda\lambda}^2 & \sigma_{\lambda\phi}^2 \\ \sigma_{\lambda\phi}^2 & \sigma_{\phi\phi}^2 \end{bmatrix} \tag{46}$$

If $\sigma_{\lambda\phi} = 0$, the semiaxes of the ellipse are oriented north-south and east-west. Because $\sigma_{\lambda\lambda}$ and $\sigma_{\phi\phi}$ were found by propagating univariate standard errors, they must be multiplied by $(2)^{1/2}$ to obtain the lengths of the semiaxes for the bivariate confidence ellipse of the pole (equation (17)). To convert this standard error ellipse to a 95% confidence ellipse, multiplication by an additional factor of 1.73 is required. The lengths of the semiaxes of the 95% confidence ellipse are thus equal to $2.45\sigma_{\lambda\lambda}$ and $2.45\sigma_{\phi\phi}$.

Early and Late Cretaceous paleomagnetic poles (Figure 13) were found using the data from analyses A, B, and C, where the data from analysis A do not include cooling units with only one measurement. These data were included in analyses B and C, assigning error estimates as previously described. Data from site 433 were not included because of the site's Paleocene age. The difference in mean poles between analyses A and C (Table 10) reflects mainly the use of different data sets in the two analyses. Comparing analyses B and C, the two-tier grouping of data can be seen to have had little effect on either the final pole or the error ellipses. For analysis C the error ellipses are several times larger than for analysis A (Figure 10), reflecting the propagation of the much larger errors of analysis C.

Histograms of the normalized residual site colatitude errors are shown in Figure 14 for analyses A and C. If the standard errors are correctly assessed, the residuals normalized by the standard errors will approximate a Gaussian distribution with standard deviation $\sigma = 1$ (curves in Figure 14). The fit between the residuals for the analysis C data and this curve is very good, whereas the residuals for the analysis A data are much too large. This is confirmed (Table 10) by chi-square tests on the residuals of all of the pole analyses. The data of analysis C (Table 11) are self-consistent, whereas the errors of analysis A are clearly too small.

The Early and Late Cretaceous poles obtained from analysis A are significantly different whereas those obtained from analysis C are not, illustrating the importance of careful estimation of errors. Since with proper error estimation the Early and Late Cretaceous poles are not significantly different, the two data sets can be combined to give a mean pole for the Cretaceous (Figure 13 and Table 10).

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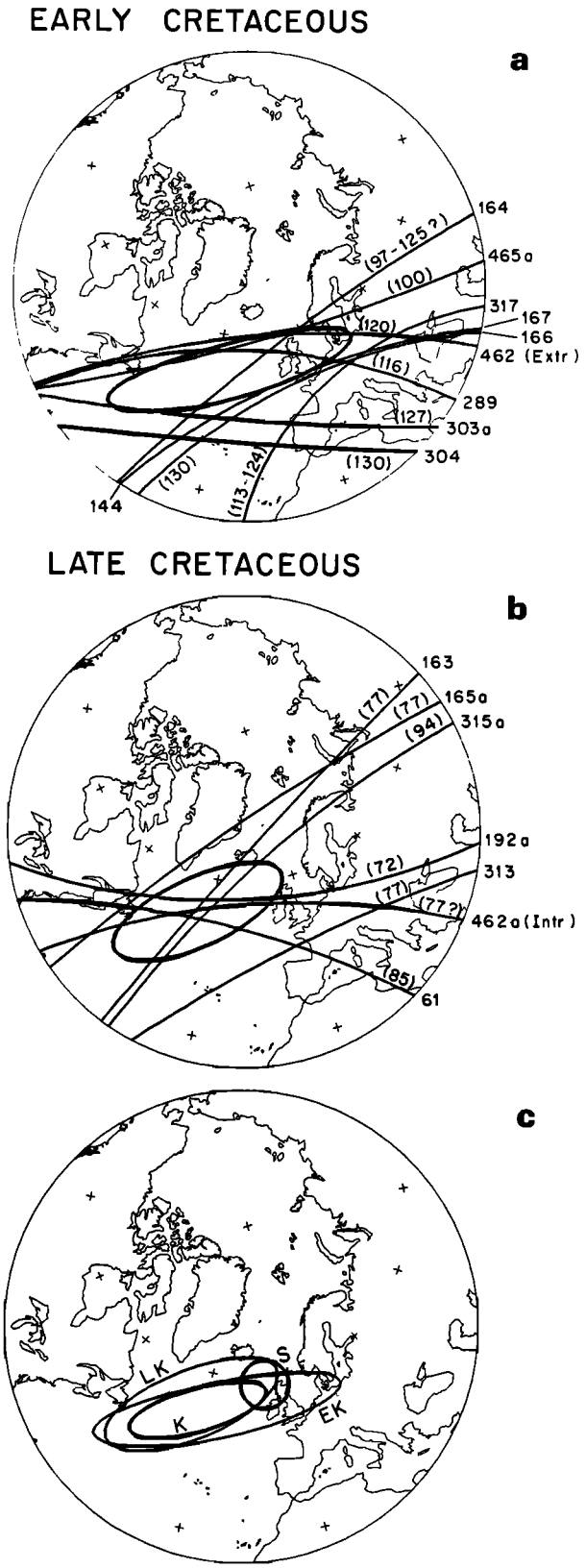


Fig. 13. Cretaceous paleocolatitude small circles and 95% confidence ellipsoids for best fit poles. Ages (in Ma) of the sampled strata are shown in parentheses along the paleocolatitude small circles. Site numbers are listed outside the circle of the equal area projection. In Figure 13*c*, EK, LK, and K are 95% confidence ellipsoids for Early Cretaceous, Late Cretaceous, and all Cretaceous data respectively; S is the mean pole from Cretaceous seamounts [*Harrison et al.*, 1975].

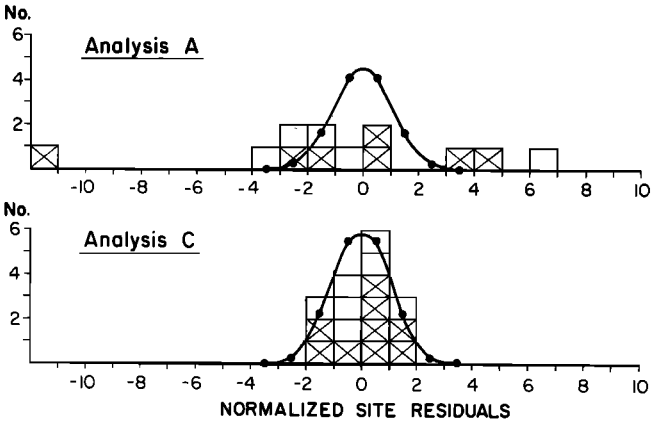


Fig. 14. Site residuals (observed mean paleocolatitude minus paleocolatitude predicted at site by best fit pole) normalized by dividing by σ, the standard error deduced from analysis a (above) and Analysis C (below). Open boxes indicate Late Cretaceous data. Boxes with diagonals indicate Early Cretaceous data. Solid circles and curves indicate values expected for normal distribution with standard deviation of one. For χ^2 values, see Table 10.

DISCUSSION

In addition to providing a test of the methods developed in this paper for analyzing paleocolatitude data, the preceding analysis provides some useful information about the paleomagnetic polar wander curve for the Pacific plate. Our Cretaceous pole at 52°N, 328°E agrees remarkably well with the pole at 50°N, 330°E determined by *Larson and Chase* [1972] from the skewness of Mesozoic magnetic lineations M-1 to M-10. Our pole is also consistent with the Cretaceous paleomagnetic pole (S in Figure 13) found [*Harrison et al.*, 1975] by analyzing the magnetic field above seamounts resting upon Mesozoic oceanic crust. The seamount pole lies on the eastern edge of the error ellipse of our paleomagnetic pole (Figure 13).

Although polar wander within the Cretaceous is not resolvable using Cretaceous borehole data alone, it is clearly apparent when additional data are considered. To detect polar wander within the Cretaceous, it is necessary to use a larger data base that includes paleomagnetic information from seamounts and the skewness of magnetic anomalies and to group these data according to age. We have done this in Figure 15, where the five small circles and poles are based on the following five separate data sets. *M* is a 28 Ma small circle based on the borehole data from Midway Island [*Grommé and Vine*, 1972]. this paleocolatitude circle establishes that at the time of the age of the basalt, 28 Ma, the site was at a colatitude of 73.1° ± 4.1°, which constrains the 28 Ma pole to lie along the small circle shown. *KT* is a 60-70 Ma paleomagnetic pole. Data used in this analysis [*Gordon*, 1982] include borehole results from site 433 basalt, paleomagnetic data from several sedimentary cores, the skewness of magnetic anomalies, and the relative amplitude of anomalies with different azimuths, all from rocks of Maastrichtian and early Paleocene age. None of these data are included in our Cretaceous pole analysis. *S* is a 80 Ma seamount pole. Data consist of four paleomagnetic poles determined from anomalies over seamounts all thought to be Campanian in age [*Gordon and Cox*, 1980]. These poles are a subset of the larger set of poles with a broader span of ages used to

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TABLE 10. Cretaceous Paleomagnetic Poles from Least Squares Analysis of Data from Table 9

Age Analysis	Early Cretaceous			Late Cretaceous		All Cretaceous
	A	B	C	A	B and C	C
Pole						
Lat, °N	46.1	52.9	53.0	52.0	53.2	52.3
Long, °E	324.4	334.5	334.7	331.6	326.0	328.1
95 % confidence ellipse						
Length of semiaxis, deg	1.0	5.4	5.4	2.0	6.7	4.2
Orientation CW from N, deg	330	340	340	326	322	330
Length of semiaxis, deg	2.6	23.7	23.6	5.5	17.5	13.7
Orientation CW from N, deg	60	70	70	56	52	60
Number M of colatitudes	7	9	9	5	7	16
Degrees of freedom	5	7	7	3	5	14
χ^2	160	7.5	7.6	66.3	3.0	11.2

find the Cretaceous seamount pole shown in Figure 13 [*Harrison et al.*, 1975]. *K* is a Cretaceous pole from borehole data. The data used in the present analysis are from rocks with ages spanning much of the Cretaceous except for the very latest part (late Maastrichtian data were included in the 60 to 70 Ma pole). *J* is the paleocolatitude small circle from site 307. This site was drilled on magnetic anomaly M21n [*Larson and Lowrie*, 1975], the age of which is probably Kimmeridgian [*Harland et al.*, 1982]. Note that the seamount pole of *Harrison et al.* [1975] and the Cretaceous pole from paleocolatitudes both are based on data spanning much of the Cretaceous and thus represent weighted average poles and not necessarily any particular instantaneous pole along the path.

The pole path in present coordinates appears to have moved slowly eastward across the Atlantic basin during the middle part of the Cretaceous, then turned and moved rapidly northward during the Late Cretaceous, and finally moved northward more slowly during the Tertiary to the present geographic pole. The general features of this path can be seen in studies by *Francheteau et al.* [1970], *Larson and Chase* [1972], *Larson and Lowrie* [1975], and *Peirce* [1976], although the broader data base now available permits the path to be determined with more certainty and resolution. Southward motion of the Pacific plate during Early Cretaceous time and possibly during latest Jurassic time, as proposed by *Larson and Lowrie* [1975] and *Peirce* [1976], rests mainly upon the results from site 307. The colatitude of this site is 82°, whereas that predicted at this site by the mean Cretaceous pole is 98.3°, the difference being $\Delta\Theta = 16.2°$. Only five flows were sampled, which we have combined to form three groups with a standard error of 5.6°. Taking into account the magnitude of the confidence ellipse for the Cretaceous pole and the standard error of the colatitude at site 307, the standard error of $\Delta\Theta$ is 6° and the 95% confidence limit is 12°. Therefore the paleocolatitude of site 307 is not consistent with the mean Cretaceous pole. Noting that this site was drilled on magnetic anomaly M21n and that the paleocolatitude from the borehole data is consistent with the anomaly skewness of anomaly M21n, *Larson and Lowrie* [1975] concluded that the age of the borehole samples is the same as that of anomaly M21n, which is Late Jurassic and probably Kimmeridgian [*Harland et al.*, 1982]. The occurrence directly above the basalt of sediments of Neocomian age leaves open the possibility that the age of the basalts is earliest Cretaceous and, as is generally true at low-latitude sites, the polarity is subject to question. However, these uncertainties in the age and polarity of the basalt of site 307 do not seriously weaken the conclusion that the Pacific plate moved toward the south before the Late Cretaceous.

The most southerly point of the pole path may well have been to the south of our mean Cretaceous pole, since the latter must include contributions from times when the pole was more northerly than it was at its southern turning

TABLE 11. All Cretaceous Paleomagnetic Data from the Pacific Plate: Analysis C

Site	Observed Colatitude	Standard Error	Model Colatitude	Residual	Importance
061	116.3	10.9	115.6	0.7	0.09
163	95.3	6.6	97.6	−2.3	0.24
164	105.3	5.8	101.5	3.8	0.19
165a	96.3	9.3	107.5	−11.2	0.07
166	125.7	9.3	115.9	9.8	0.05
167	121.5	7.7	113.5	8.0	0.06
192a	70.5	7.4	73.8	−3.3	0.11
289	123.3	11.5	127.5	−4.2	0.04
303a	95.8	9.2	86.8	9.0	0.10
304	101.4	9.6	88.2	13.2	0.08
313	112.3	11.0	99.3	13.0	0.03
315a	107.4	5.3	107.8	−0.4	0.32
317a	140.9	6.6	137.2	3.7	0.08
462a(extr.)	112.8	5.5	112.8	−6.0	0.13
462a(intr.)	117.3	4.6	118.8	−1.5	0.19
465a	86.3	3.8	89.8	−3.5	0.22

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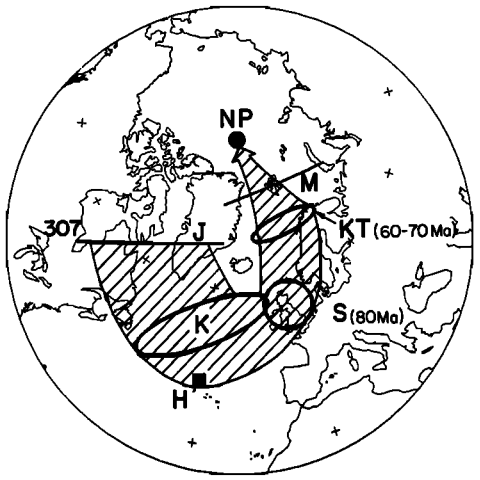


Fig. 15. Pacific plate paleomagnetic polar wander curve. Labels are as follows: NP, present north pole; M, paleocolatitude small circle determined from Midway Island (28 Ma) by *Grommé and Vine* [1972]; KT, pole for 60 to 70 Ma [*Gordon*, 1982] based on a variety of paleomagnetic results from rocks of Maastrichtian and possibly early Paleocene age; S, pole from anomalies over four seamounts thought to be ~80 m.y. in age [*Gordon and Cox*, 1980]; K, mean Cretaceous pole (this study) excluding late Maastrichtian data; H, pole for four sites (166, 167, 303a, and 304) thought to be Hauterivian (131 to 125 Ma) or older with coordinates 41°N and 329°E; J, paleocolatitude circle of site 307 drilled on seafloor beneath anomaly M21n [*Larson and Lowrie*, 1975], the age of which [*Harland et al.*, 1982] is probably Kimmeridgian (156 to 150 Ma).

point. A turning point well to the south is suggested by the southerly location of the colatitude circles for the four sites with ages of Hauterivian or older (sites 166, 167, 303a, and 304). Analysis of these four data yields a pole (H in Figure 14) at 41°N, 329°E. (The 95% confidence ellipse has major and minor axes of 33° and 11°, the orientation of the major axis is 64° clockwise from north; the chi-square statistic with 2 degrees of freedom is 0.1.) Comparison of this pole with the colatitude circle for site 307 indicates a southward displacement of the pole of 27° ± 8°, during a time no longer than that between the Kimmeridgian and the Hauterivian ages, roughly 25 m.y. The indicated rate of southward motion of the Pacific plate is one degree per million years, close to the maximum rate of present plate motion.

In summary, our present analysis supports the view of *Larson and Lowrie* [1975] and *Peirce* [1976] that during the earliest Cretaceous and possibly latest Jurassic time the Pacific plate and its pole were moving rapidly toward the south. During mid-Cretaceous time the pole moved in an easterly direction at a much slower rate. It then turned and moved in a northerly direction at a rapid rate during the Late Cretaceous, corresponding to rapid northward motion of the Pacific plate. During the Tertiary the plate decelerated and continued slowly northward to its present position.

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(Received May 11, 1983;
accepted August 1, 1983.)

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